

Moving rational sources, arithmetic collisions and complete observed memory

Split-Zero research programme

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The original arithmetic observation now has a kernel-energy bound valid throughout source activation. This bound controls the complete observed memory at its late spectral scale, where the earlier comparison for subexponential regularizers did not apply. The mass in this receiver is the actual derivative of its Schur pencil:

$$A_N \succeq a_N I, \quad W_N = \mathcal{F}'_N(0) = I + B_N^\dagger A_N^{-2} B_N.$$

The proof retains every coupling block. In the original four-cutoff return, at the specified critical source activation and native late regularizer, it gives

$$\mathcal{R} \log \frac{\det \mathcal{F}_N(z_k^*)}{\det \mathcal{F}_N(0)} = 2q \log(4/\pi) + o(q).$$

At the canonical endpoint the same quantity is $4q \log(4/\pi) + o(q)$ (KG39–42). The mass-weighted full observed heat and the explicit energy-graph heat both have return two in the strict critical interval. The raw observed weights remain in their own exact formulas.

A second result allows all rational poles to move and collide inside a fixed compact set disjoint from the original arithmetic lattice. The source and observed Grams have the same harmonic activation scale; every least filter singular value is determined up to $e^{o(q)}$. The full observed heat operator is localized in the observed rational sector with a quantitative trace-norm error (MF9, MF34–41).

At collisions with the arithmetic spectrum itself, a polynomial frame continues the entire sector. PC1–10 evaluates its resultant and its complete primary kernel. PC15–34 gives corrected chain sections, retains any new observation kernel, and proves its exact determinant transfer. Every source coordinate, original minimum and kernel factor is retained. These finite boundary formulas extend the maps; they do not import an off-spectrum analytic bound across a vanishing observed fraction.

The first canonical kernel allocation and the original complex-current phases still require their own evaluations. The complete proofs below state the finite source guards and the exact retained status of the high-endpoint analytic inputs. Source files and earlier complete providers accompany this reader. Published edition 016 is preserved.

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Two different effects at the same arithmetic collision

For the exact auxiliary algebra $Q = y^2 + 1$, take $D = y - z$ and $z = i + t$, with the real parameter t shown below. The coefficient metric is $G = I$ only in this stated example. The globally defined frame is $F_D = (z, 1)^\top$.

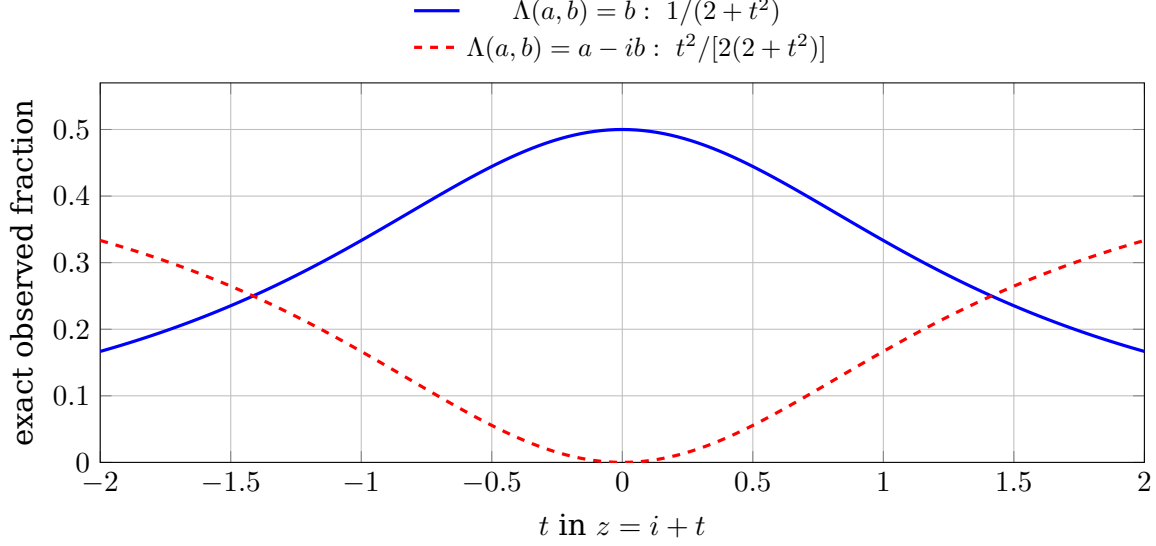


Figure 1: PC29–30 gives both exact curves from the full quotient metric $Q_B = (\Lambda\Lambda^*)^{-1}$. At $t = 0$ the rational numerator inverse has a pole in both cases. The first observation continues to detect the sector. The second acquires the one-dimensional kernel $Z = \mathbb{C}(i, 1)^\top$, retained by PC25 and PC27. This is an exact finite illustration, not a computed zeta-zero packet.

$$\begin{array}{c}
 \boxed{E \xrightarrow{\Lambda} B} \\
 \downarrow p \qquad \uparrow j \\
 \boxed{Z \oplus (E/V) \xrightarrow{(z, c) \mapsto \bar{\Lambda}c} B/W} \\
 \\
 Z = V \cap \ker \Lambda, \quad W = \Lambda V \\
 pj = I, \quad I - jp = dh + hd
 \end{array}$$

Figure 2: The exact reduction retains the new kernel Z when observation rank drops. The minimum source section is adjusted by $(I - h\Lambda)$; PC17–28 proves every arrow and its metric. PC32–34 gives the positive-determinant transfer on K/Z and retains the inherited metric on Z .

The full memory and its actual mass

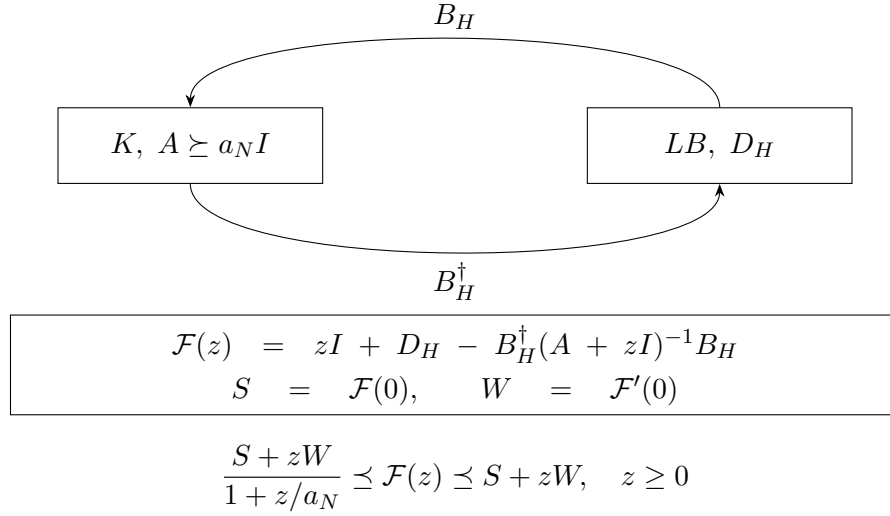


Figure 3: KG17–30 retains the full arithmetic energy blocks and proves the displayed finite relative estimate. The mass W is the first time moment of the original memory, plus the identity. The Schur construction is connected to the original-author treatment of Dusson, Sigal and Stamm, *The Feshbach-Schur map and perturbation theory* (2021), theorem `thm:isospF` and equation `QP`, arXiv:2105.02058v1. The activated arithmetic estimates and their complete proofs are KG1–42.

Part I

Kernel energy and complete observed memory

The original kernel gap, arithmetic powers, and complete late memory

Independent derivation, 21 September 2026. Equations **KG1-KG42** are the locators in this proof. The complete received source, equations (1)-(36), is retained as SUPPLIED_KERNEL_GAP.md. This proof verifies its finite claims, strengthens several bounds, and specifies the scope of its critical limits. No metric, source mass, coordinate, period coefficient, or relation is replaced.

1. Original objects, finite source constants, and exact domain

Retain the simple quartet

$$q = (k+1)^2, \quad k \geq 9, \quad k \equiv 1 \pmod{4}, \quad 0 < \delta < 1/2, \quad \gamma > 2, \quad R_0 = \sqrt{\delta^2 + \gamma^2},$$

$$S = k/2 + iy, \quad Q(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \quad E = \mathbb{C}[y]/(Q), \quad M[p] = [yp]. \quad (\text{KG1})$$

The source is the original $d\mu_k = w_h^{*k}(y) dy$, with $w_h = |(2\xi/h)(1/2 + iy)|^2/(2\pi)$. The whole divisor, source mass and polynomial relation space are retained. At $q-1 \leq N \leq 2q$, write $G_0 = G_N$ for the full canonical minimum and $G_1 = G_N^J$ for the complete joint-source minimum. Their exact activation is

$$G_\alpha^{-1} = (1-\alpha)G_0^{-1} + \alpha G_1^{-1}, \quad 0 \leq \alpha \leq 1. \quad (\text{KG2})$$

Let $\Lambda : E \rightarrow B$ be the unchanged onto observation, $K = \ker \Lambda$, and

$$Q_B = (\Lambda G_\alpha^{-1} \Lambda^*)^{-1}, \quad L = G_\alpha^{-1} \Lambda^* Q_B, \quad P_B = \Lambda L. \quad (\text{KG3})$$

Multiplication gives $\Lambda L = I$, $L^* G_\alpha L = Q_B$, $P_B^2 = P_B = P_B^\dagger$, and $L^\dagger = \Lambda$. Thus L is the exact minimum-lift isometry to $K^\perp_{G_\alpha}$. The fixed original coefficient frame is denoted $I_K : \mathbb{C}^m \rightarrow E$, its inherited Gram is $K_0 = I_K^* G_\alpha I_K$, and its isometric version is $J_K = I_K K_0^{-1/2}$. In an intrinsic statement on K , the inclusion is an isometry with its inherited metric; in the coefficient frame, the identical gap statement is a comparison with K_0 , not with an unrelated identity matrix.

On the original proved five-orbit period locus, $m = 8k - 16$, $n = q - m = k^2 - 6k + 17$. The complete frame is $I_K = V^{-1} \pi^T$, with $\ker \pi = \text{im}[M_A, S_k Z_k]$; the transpose is complex-linear duality. These are MR1-8. Its verified period domain, including $|u| \geq R_*$, is recorded in NG24. All asymptotic rank uses below retain that domain; none assigns the five-orbit rank on a size-one orbit.

Here are the precise source bounds used. Put $U_s d = \sum_{j=1}^s d_j [y^{-j}]$, $V_s = \text{im } U_s$. The proved all-activation comparison HAR7-12, HAR26-31 gives

$$U_s^* \Lambda^* Q_B \Lambda U_s \succeq \ell_{s,N} U_s^* G_\alpha U_s, \quad \ell_{s,N} = e^{-2E_{s,N}} \in (0, 1],$$

$$E_{s,N} = O_{h,\omega}(k \log^2(q+2) + (k+Js) \log(q+Js+2)). \quad (\text{KG4})$$

Here J is the number of actual nonzero conductor coefficients; ϖ denotes the fixed original period data, not a freely selected replacement. For a finite value use exactly $E_{s,N} = \max(0, -\log L_{s,N}, \log U_{s,N})$, $L_{s,N} = a_{s,N} / \max(1, \mathbf{L}_k \Gamma_N)$, $U_{s,N} = b_{s,N} + \mathbf{B}_k A_w^k / (R_0^2 - 1)$, with $a_{s,N}, b_{s,N}$ the original finite IVO constants, Γ_N the explicit HAR20 cardinal constant, and A_w^k the actual total tensor weight. HAR10 and HAR19-26 define all these finite quantities; no limiting constant is used to stand in for them.

The finite guards, HAR11, are

$$\begin{aligned} 1 \leq s \leq q, \quad q_- = (k-7)^2 \geq 16, \quad kR_0 \leq 2^{-14} q_-, \\ n_- + 2 \leq 2q_-, \quad \mathcal{L}_* \mathcal{R}_* T_N^{\text{lb}} \geq 2\mathcal{A}, \quad n_- = N - v + Js - q_-. \end{aligned} \quad (\text{KG5})$$

The last inequality uses an actual nonzero selected conductor coefficient and its complete pole numerator, exactly as in IVO24; all original invariant constraints remain. IVO24-26 proves these guards eventually for fixed original data and $s \log(q+2) = o(q)$. Small algebraic fixtures below do not assert these eventual guards.

The full-polynomial canonical comparison is $\ell_{k,N} \|p\|_\sigma^2 \leq \|p\|_{\mu_k}^2 \leq u_k \|p\|_\sigma^2$, with the original Gamma mass $\sqrt{2\pi}$. Anchored division on polynomials through degree N has Gamma norm at most $\mathfrak{d}_N = 2\sqrt{2N+1}(\sum_{j=1}^N 1/j)(1 + \sqrt{N+1})$. The joint metric satisfies $\mathbf{L}_k^{-1} T_k \preceq G_1 \preceq \mathbf{B}_k T_k$, with T_k diagonal in the exact original root-value coordinates. These statements and their proofs are ACT3-9 and HAR5-6. Their logarithmic losses are $O_{h,\varpi}(k \log^2(q+2))$. The Gamma recurrence and source comparison retain the human-source attribution in those proofs; no new analytic-source asymptotic is claimed here.

2. Exact inverse transport and a sharp kernel gap

Let $\hat{R}_N : E \rightarrow \mathcal{P}_N$ be the complete canonical minimum section, J_N the remainder map, $\mathscr{D}p = (p - p(0))/y$, and $u = [y^{-1}]$. All roots of Q have modulus at least $R_0 > 2$, so M and this class are defined. Multiplication modulo the full Q proves

$$M^{-1} = B_{1,0} + u\lambda_{1,0}, \quad B_{1,0} = J_N \mathscr{D} \hat{R}_N, \quad \lambda_{1,0}(x) = (\hat{R}_N x)(0). \quad (\text{KG6})$$

In particular $\lambda_{1,0}$ evaluates the specified lift, not a purported quotient evaluation at zero. For general $1 \leq s \leq q$, writing a polynomial as its first s Taylor coefficients plus $y^s \mathscr{D}^s p$ proves

$$M^{-s} = B_{s,0} + U_s R_{s,0}, \quad B_{s,0} = J_N \mathscr{D}^s \hat{R}_N, \quad (R_{s,0} x)_j = [y^{s-j}] \hat{R}_N x, \quad 1 \leq j \leq s. \quad (\text{KG7})$$

Every division is applied to a polynomial at the original cutoff; no degree-increased source is introduced.

Define the actual coisometry $\mathcal{C}_\alpha(x_0, x_1) = \sqrt{1-\alpha}x_0 + \sqrt{\alpha}x_1$ from $E_{G_0} \oplus E_{G_1}$ to E_{G_α} . Its adjoint is $(\sqrt{1-\alpha}G_0^{-1}G_\alpha x, \sqrt{\alpha}G_1^{-1}G_\alpha x)$, and KG2 gives $\mathcal{C}_\alpha \mathcal{C}_\alpha^\dagger = I$. For any fixed coefficient operator T , $T = \mathcal{C}_\alpha \text{diag}(T, T) \mathcal{C}_\alpha^\dagger$, so $\|T\|_{G_\alpha} \leq \max(\|T\|_{G_0}, \|T\|_{G_1})$. Applying this identity to KG7 yields exactly

$$\begin{aligned} M^{-s} &= B_{s,\alpha} + U_s R_{s,\alpha}, \\ B_{s,\alpha} &= (1-\alpha)B_{s,0}G_0^{-1}G_\alpha + \alpha M^{-s}G_1^{-1}G_\alpha, \\ R_{s,\alpha} &= (1-\alpha)R_{s,0}G_0^{-1}G_\alpha, \\ \|B_{s,\alpha}\| &\leq D_s := \max \left\{ 1, \mathfrak{d}_N^s \sqrt{u_k / \ell_{k,N}}, \sqrt{\mathbf{L}_k \mathbf{B}_k} R_0^{-s} \right\}. \end{aligned} \quad (\text{KG8})$$

The first bound follows because J_N is a contraction and $\widehat{R}_N$ is an isometry, applying source comparison before and after s polynomial divisions. In T_k , the operator M^{-s} is diagonal, of norm R_0^{-s} . The joint sandwich gives the second bound. The norm of the coisometry and its adjoint is one. Thus KG8 includes both endpoints $\alpha = 0, 1$, with the exact factor order and rank at most s .

Here is the underlying subspace statement. If $V \subset E$ satisfies $\|P_B v\|^2 \geq \ell \|v\|^2$ for all $v \in V$, then $\|P_K|_V\| \leq \sqrt{1-\ell}$. Its adjoint is $P_V|_K$, so the same bound holds there. Orthogonal decomposition gives

$$\|(I - P_V)x\| \geq \sqrt{\ell}\|x\| \quad (x \in K). \quad (\text{KG9})$$

Apply this to V_s using KG4. Since $(I - P_{V_s})U_s = 0$, applying KG8 to $M^s x$ gives $(I - P_{V_s})x = (I - P_{V_s})B_{s,\alpha}M^s x$. Hence

$$\|M^s x\| \geq e^{-E_{s,N}} D_s^{-1} \|x\| \quad (x \in K). \quad (\text{KG10})$$

For $s = 1$, put $D_- = D_1$, $\ell_N = \ell_{1,N}$, $a_N = \ell_N/D_-^2$. In the original metric,

$$J_K^\dagger M^\dagger M J_K \succeq a_N I_m, \quad I_K^* M^* G_\alpha M I_K \succeq a_N K_0, \quad -\log a_N = O_{h,\varpi}(k \log^2(q+2)). \quad (\text{KG11})$$

The full action is used. In its original orthogonal blocks the left side is $M_K^\dagger M_K + C_0^\dagger C_0$, including the leakage $C_0 : K \rightarrow B$. No inequality for M_K alone follows. For instance the invertible swap matrix, with $K = \mathbb{C}e_1$, has $M_K = 0$ but $M^\dagger M = I$.

The constant ℓ/D_-^2 is sharp with only the given angle and residual data. In a declared Euclidean fixture, let $u = e_1$, $K = \mathbb{R}(4/5, 3/5)$, $M^{-1} = \begin{pmatrix} 2 & 4/3 \\ 0 & 1 \end{pmatrix} = \text{diag}(0, 1) + u(2, 4/3)$. Here $D_- = 1$, the observed fraction of u is $\ell = 9/25$, and $M(4/5, 3/5) = (0, 3/5)$; equality holds in KG11. This fixture also shows why no separation guard is necessary for the proof.

KG11 strengthens the canonical singular-line guard in ME27-29 and extends its gap to every activation. Its finite requirements are exactly the original harmonic guards KG5, not a new assumed gap. The full quotient's scale $\exp[-2q \log(q/k) + O(q)]$ is a canonical or fixed/bounded critical-parameter assertion. It is not uniform over $0 \leq \alpha \leq 1$: at $\alpha = 1$, $\|M^{-1}\| \leq \sqrt{\mathbf{L}_k \mathbf{B}_k}/R_0 = e^{o(q)}$. The kernel gap itself remains uniform there.

3. Complete word volumes in the original coefficient frame

The original outgoing decomposition is $M = C_N + r_N \ell_N^{\text{top}}$, with $\|C_N\| \leq C_{\text{pol}}(N)$, $\|r_N \ell_N^{\text{top}}\| = \varepsilon_N$; it is selfadjoint only in G_0 . Transporting its bounded and rank-one parts by the coisometry proves

$$s_1(M; G_\alpha) \leq L_+, \quad s_j(M; G_\alpha) \leq D_+ \quad (j \geq 2),$$

$$D_+ = \max\{1, C_{\text{pol}}(N), \sqrt{\mathbf{L}_k \mathbf{B}_k} k R_0\}, \quad L_+ = \max\{D_+, C_{\text{pol}}(N) + \varepsilon_N\}. \quad (\text{KG12})$$

Indeed the transported bounded part has norm at most D_+ and the remaining term still has rank at most one. On its row kernel, singular min-max gives $s_2 \leq D_+$. For s_1 , apply the coisometry to the complete M ; its canonical norm is at most $C_{\text{pol}} + \varepsilon_N$, and its joint norm is at most $\sqrt{\mathbf{L}_k \mathbf{B}_k} k R_0$. Thus the stated maximum, rather than an unproved sum estimate, is valid. The original profile H5 and source constants give $\log L_+ = O_{h,\varpi}(q)$, $\log D_+ = O_{h,\varpi}(k \log^2(q+2))$. No selfadjointness of the transported bounded part is asserted.

Let $H = M^\dagger M$, $K_s^M = (M^s I_K)^* G_\alpha(M^s I_K)$, and $K_j^H = (H^j I_K)^* G_\alpha(H^j I_K)$. Put $D_* := 2 \log L_+ + 2(m-1) \log D_+$. Then

$$-2m(E_{s,N} + \log D_s) \leq \log \frac{\det K_s^M}{\det K_0} \leq s D_*. \quad (\text{KG13})$$

The lower bound follows from KG10 by positive congruence. For the upper bound, in orthonormal singular frames $\|\wedge^m M\| = \prod_{i=1}^m s_i(M) \leq L_+ D_+^{m-1}$. The identity $\wedge^m(M^s) = (\wedge^m M)^s$ and submultiplicativity give the upper bound. The squared norm of $\wedge^m(M^s J_K)$ is exactly $\det K_s^M / \det K_0$, so the fixed coefficient frame has not been changed.

For a unit $x \in K$, write its spectral weights for H as p_i . Convexity of $t \mapsto t^{2j}$ on $[0, \infty)$, and KG11, imply $\langle x, H^{2j} x \rangle = \sum p_i \lambda_i^{2j} \geq (\sum p_i \lambda_i)^{2j} \geq a_N^{2j}$. For the upper bound the singular values of H^j are $s_i(M)^{2j}$. Thus

$$2mj \log a_N \leq \log \frac{\det K_j^H}{\det K_0} \leq 2j D_*. \quad (\text{KG14})$$

Equations (8) and (9) of the arrival are KG13 at $s = 1$ and KG14 at $j = 1$. Their K_1 is K_1^M and their K_2 is K_1^H ; this distinction prevents confusing H with M^2 .

More generally, for any positive integer bound r_k such that

$$\frac{r_k \log^2(q+2)}{k} \rightarrow 0, \quad (\text{KG15})$$

KG13-14 are $o(kq)$, uniformly for $1 \leq s, j \leq r_k$, in N and α . To verify each term, $m = O(k)$, $q \asymp k^2$, $D_* = O(q + k^2 \log^2 q)$, $\log \mathfrak{d}_N = O(\log q)$, and $\log D_s = O(k \log^2 q + s \log q)$. Since KG15 implies $s = o(k)$, KG4 gives $E_{s,N} = O(k \log^2 q)$. The lower side of KG13 is $O(k^2 \log^2 q + ks \log q) = o(kq)$; its upper side and both sides of KG14 are $O(r_k(q + k^2 \log^2 q)) = o(kq)$. Also $r_k \log q = o(q)$, so every finite source guard is eventually satisfied. The received choice $\max(1, \lfloor k / \log^3(q+2) \rfloor)$ satisfies KG15.

Taking the four original signs therefore proves that the returns of K_s^M and K_j^H differ from the return of K_0 by $o(kq)$. The estimate is uniform even if the four activations are chosen independently. It concerns precisely these two families of words; no bound for every possible noncommuting word is inferred.

There is an exact quotient action. For any invertible original coefficient operator T , use ΛT^{-1} as coordinates on E/TK . Then $\bar{T} : E/K \rightarrow E/TK$ has coordinate matrix identity, with source metric Q_B and target metric $Q_T = (\Lambda T^{-1} G_\alpha^{-1} T^{-*} \Lambda^*)^{-1}$. Two full Gram eliminations, or completion of the energy square on the original kernel, give

$$\log \det(\bar{T}^\dagger \bar{T}) = \log \frac{\det Q_T}{\det Q_B} = 2 \log |\det T| - \log \frac{\det(I_K^* T^* G_\alpha T I_K)}{\det K_0}. \quad (\text{KG16})$$

For detail, choose any complement frame F with $\Lambda F = I$. The determinant of $[I_K, F]^* G [I_K, F]$ is $\det K_0 \det Q_B$. Replacing G by $T^* G T$ changes its determinant by $|\det T|^2$, its kernel factor to the one in KG16, and its quotient factor to Q_T . Cancel the same frame determinant. For $T = M^s$, the first term is $2s \log |Q(0)|$, independent of cutoff, and cancels under the four signs. For $T = H^j$ it is $4j \log |Q(0)|$, again independent of cutoff although the adjoint in H depends on the metric. Both induced quotient actions have zero leading kq return in KG15. The map is not $(\Lambda M L)^s$, whose intermediate projections are absent from this exact quotient construction.

4. Complete energy and the derivative mass

Use $K \oplus LB$, with its inherited metrics, and write

$$H = \begin{pmatrix} A & B_H \\ B_H^\dagger & D_H \end{pmatrix} > 0, \quad A \succeq a_N I, \quad F(z) = zI + D_H - B_H^\dagger(A + zI)^{-1}B_H. \quad (\text{KG17})$$

Here $A = M_K^\dagger M_K + C_0^\dagger C_0$, $B_H = M_K^\dagger D_0 + C_0^\dagger M_B$, $D_H = D_0^\dagger D_0 + M_B^\dagger M_B$. Every coupling block is retained. Block elimination gives $F(z)^{-1} = \Lambda(H + zI)^{-1}L$ whenever the inverses exist.

The general Schur construction and reconstruction map are presented by Geneviève Dusson, Israel Michael Sigal and Benjamin Stamm, *The Feshbach-Schur map and perturbation theory* (2021), original author TeX, arXiv:2105.02058v1, theorem thm:isospF, reconstruction equation QP, and proposition prop:U-prop-SA. Its intact original source was retained for RM18-26 in the preceding edition; its complete source lines 398-507 were reread for the present proof. The identities and activated finite estimates KG17-38 are proved here; the human source is credited for the general construction, not asserted to supply these new constants or their original arithmetic receivers.

Put

$$S = F(0) = D_H - B_H^\dagger A^{-1} B_H > 0, \quad W = F'(0) = I + B_H^\dagger A^{-2} B_H \succeq I. \quad (\text{KG18})$$

With $\mathcal{E} = L - J_K A^{-1} B_H$, where B_H is the map from the original observed Hilbert space to the isometric kernel, direct multiplication and completion of the quadratic square prove $\Lambda \mathcal{E} = I$, $\mathcal{E}^\dagger \mathcal{E} = W$, $\mathcal{E}^\dagger H \mathcal{E} = S$. Thus

$$\mathcal{J} = \mathcal{E} W^{-1/2}, \quad \mathcal{J}^\dagger \mathcal{J} = I, \quad H_{\text{eff}} = \mathcal{J}^\dagger H \mathcal{J} = W^{-1/2} S W^{-1/2}. \quad (\text{KG19})$$

These are identities between the original observed metric and the original full metric. In arbitrary observed coefficient coordinates, the intrinsic endomorphisms S, W have coefficient Grams $Q_B S, Q_B W$, and their adjoints use Q_B . Under the exact isometry $b \mapsto Q_B^{1/2} b$, their Euclidean Hermitian matrices are $S_o = Q_B^{1/2} S Q_B^{-1/2}$, $W_o = Q_B^{1/2} W Q_B^{-1/2}$. The same coefficient Grams are consequently $Q_B^{1/2} S_o Q_B^{1/2}$ and $Q_B^{1/2} W_o Q_B^{1/2}$. The intrinsic positive square root of W is transported from that of W_o . These formulas record every coordinate factor in the incoming inclusion notation.

Let $K_1 = K_1^M$, $K_2 = K_1^H$. Since the kernel block of H^2 is $A^2 + B_H B_H^\dagger$, the rectangular determinant identity gives

$$\boxed{\det W = \frac{\det K_2 \det K_0}{(\det K_1)^2}, \quad \det S = |Q(0)|^2 \frac{\det K_0}{\det K_1}, \quad \det H_{\text{eff}} = |Q(0)|^2 \frac{\det K_1}{\det K_2}.} \quad (\text{KG20})$$

Indeed $\det A = \det K_1 / \det K_0$, $\det(J_K^\dagger H^2 J_K) = \det K_2 / \det K_0$, and $\det W = \det(A^2 + B_H B_H^\dagger) / (\det A)^2$. The identity $\det(I + UV) = \det(I + VU)$ follows by eliminating either diagonal block of $\begin{pmatrix} I & U \\ -V & I \end{pmatrix}$. Full elimination gives $\det H = \det A \det S = |\det M|^2 = |Q(0)|^2$. These prove every factor in KG20, agreeing with ME30-33 and BR1-6.

A stronger bound than incoming (16) follows from the exact isometry $V = H^{1/2} J_K A^{-1/2}$. Multiplication gives $V^\dagger V = I$ and $\det(V^\dagger H V) = \det A \det W$. For any m -dimensional isometry J , singular min-max and KG12 give $\det(J^\dagger H J) \leq L_+^2 D_+^{2(m-1)}$. Hence

$$\boxed{0 \leq \log \det W \leq D_* - \log \det A \leq D_* + m \log(1/a_N) = O_{h,\varpi}(q + k^2 \log^2(q+2)) = o(kq).} \quad (\text{KG21})$$

For completeness, incoming (16) is valid: H equals a rank-one top-eigenvalue term plus a positive remainder of norm at most D_+^2 ; so its cross block has a rank-one part and remainder of norm at most D_+^2 . Left multiplication by A^{-1} bounds its second and later singular values by D_+^2/a_N , and its first by L_+^2/a_N . Only $\rho = \text{rank } B_H$ values are nonzero, giving exactly that expression with $\rho = 0$ interpreted as zero. Positivity also yields the sharper norm bound

$$\boxed{\|W\| \leq 1 + L_+^2/a_N.} \quad (\text{KG22})$$

To prove it, $B_H^\dagger A^{-1} B_H \preceq D_H$, while $A^{-2} \preceq a_N^{-1} A^{-1}$. Therefore $W - I \preceq a_N^{-1} D_H \preceq (L_+^2/a_N)I$. The received larger bound $1 + L_+^2/a_N^2$ remains valid.

Every restriction and attained quotient of the mass metric has cost between zero and $\log \det W$, relative to the corresponding restriction or quotient of its original base metric. For a restriction, pass by its exact base-metric isometry to $J^\dagger W J$. Its eigenvalues are at least one and bounded by the corresponding leading eigenvalues of W ; their logarithmic sum is at most $\log \det W$. For an attained quotient, use the same kernel-complement Gram elimination as in KG16: the full cost is the sum of the nonnegative kernel restriction cost and quotient cost. Iteration covers a restriction followed by a quotient. This proves the statement with the required base metrics and fixed coefficient maps, not an absolute determinant bound for arbitrarily rescaled frames.

KG13-14 and KG20-21 give $\mathcal{R} \log \det S = o(kq)$, $\mathcal{R} \log \det W = o(kq)$, and $\mathcal{R} \log \det H_{\text{eff}} = o(kq)$, uniformly in activation. The kernel's initial kq coefficient is not assigned by these distortion identities.

5. All-activation memory down to zero

Define the full memory logarithm $g(z) = \log \det(zI + A) + \log \det(zI + D_H) - \log \det(zI + H)$. Since $H > 0$, all three matrices remain positive at zero. With $\mathcal{Q}(t) = \text{Tr } e^{-tH} - \text{Tr } e^{-tA} - \text{Tr } e^{-tD_H}$, scalar Jensen in concatenated eigenbases of A, D_H gives $\mathcal{Q}(t) \geq 0$. Taylor expansion gives $\mathcal{Q}(t) = t^2 \|B_H\|_{\text{HS}}^2 + O(t^3)$, while the finite positive matrices give exponential decay at infinity. Pairing their scalar eigenvalue integrals proves $g(z) = \int_0^\infty e^{-zt} \mathcal{Q}(t) dt/t$, including zero, and $0 \leq g(z) \leq g(0)$.

Let $L_- \geq \|M^{-1}\|$, $L_- \geq D_-$, with every remaining singular value of M^{-1} at most D_- , as KG8 at $s = 1$ proves. Eliminating D_H gives $g(0) = \log \det A + \log \det((H^{-1})_{KK})$. Restriction min-max for H and H^{-1} therefore proves

$$\boxed{0 \leq g(z) \leq g(0) \leq 2 \log(L_+ L_-) + 2(m-1) \log(D_+ D_-), \quad z \geq 0.} \quad (\text{KG23})$$

One finite all-activation choice is $L_- = \max(D_-, \|M^{-1}\|_{G_0}, \|M^{-1}\|_{G_1})$; the coisometry proves its validity. ACT10-14 and the retained scalar endpoints give $\log L_- \leq q \log(q/k) + O_{h,\varpi}(q)$, uniformly in N . Consequently

$$\boxed{\sup_{N, \alpha \in [0,1], z \geq 0} g_{N,\alpha}(z) = O_{h,\varpi}(q \log(q/k) + k^2 \log^2(q+2)) = o(kq).} \quad (\text{KG24})$$

This is an additional all-activation extension of the canonical bound ME20. It uses both actual inverse and outgoing decompositions; it does not estimate q bounded singular directions by D_+ . It keeps the entire heat-memory integral at zero. It

does not delete the separate leakage term of ME21-26 or turn the independently compressed M_B into the complete Schur action.

6. Relative low-frequency comparison and every memory moment

The scalar resolvent identity on the positive kernel block, followed by the full B_H -congruence, gives exactly

$$F(z) = S + zW - z^2 B_H^\dagger A^{-2} (A + zI)^{-1} B_H. \quad (\text{KG25})$$

For real $z \geq 0$, write it instead as $F(z) = S + zI + zB_H^\dagger A^{-2} [A(A + zI)^{-1}] B_H$. The bracket lies between $(1 + z/a_N)^{-1}I$ and I , so

$$S + zI + \frac{z}{1 + z/a_N} (W - I) \preceq F(z) \preceq S + zW. \quad (\text{KG26})$$

The left side is at least $(S + zW)/(1 + z/a_N)$, proving incoming (20) with a stronger intermediate lower bound. The nonnegative difference $S + zW - F(z)$ has rank at most ρ . Congruence by $(S + zW)^{-1/2}$ shows that at most ρ eigenvalues differ from one and that each lies in $[(1 + z/a_N)^{-1}, 1]$. Thus

$$0 \leq \log \det(S + zW) - \log \det F(z) \leq \rho \log(1 + z/a_N). \quad (\text{KG27})$$

Inverting KG26 and conjugating by $W^{1/2}$ proves exactly

$$(zI + H_{\text{eff}})^{-1} \preceq W^{1/2} \Lambda (zI + H)^{-1} L W^{1/2} \preceq (1 + z/a_N) (zI + H_{\text{eff}})^{-1}. \quad (\text{KG28})$$

All these finite estimates include $z = 0$. The heat measurements derived below use this actual mass rather than treating the pencil as $zI + S$.

For every integer $p \geq 1$ and complex $|z| < a_N$, the finite geometric identity gives

$$F(z) = S + zW + \sum_{j=2}^p (-1)^{j+1} z^j B_H^\dagger A^{-(j+1)} B_H + R_p(z),$$

$$R_p(z) = (-1)^p z^{p+1} B_H^\dagger A^{-(p+1)} (A + zI)^{-1} B_H, \quad \|W^{-1/2} R_p(z) W^{-1/2}\| \leq \frac{|z|^{p+1}}{a_N^p (1 - |z|/a_N)}. \quad (\text{KG29})$$

For the norm bound, move one factor A^{-1} to each side of the middle kernel operator. Its norm is at most $a_N^{-(p-1)}/(a_N - |z|)$, while $W^{-1/2} B_H^\dagger A^{-2} B_H W^{-1/2} = I - W^{-1} \preceq I$. This proves the full complex remainder without commuting B_H with A .

Diagonalizing only A proves for the exact memory $\mathcal{M}(t) = B_H^\dagger e^{-tA} B_H$ and every integer $j \geq 0$,

$$\int_0^\infty t^j \mathcal{M}(t) dt = j! B_H^\dagger A^{-(j+1)} B_H, \quad \int_T^\infty \mathcal{M}(t) dt \preceq e^{-a_N T} B_H^\dagger A^{-1} B_H. \quad (\text{KG30})$$

The scalar integral follows by repeated integration by parts. Thus $W - I$ is the first time moment. The pencil's kernel poles are at negatives of eigenvalues of A , all of magnitude at least a_N ; the much smaller full-system pole, when present, arises from the zero of the complete Schur pencil, not an omitted kernel eigenvalue.

7. Exact soft-mode mass, energy and heat

The rank-one inverse decomposition implies $\lambda_2(H) \geq D_-^{-2}$. Let $\lambda = \lambda_1(H)$ and retain the explicit guard

$$\lambda < a_N = \ell_N/D_-^2 \leq D_-^{-2}. \quad (\text{KG31})$$

This makes the smallest eigenvalue simple. Write its unit eigenvector as $\psi = (x, b) \in K \oplus LB$, and $\theta = \|b\|_{Q_B}^2$. Necessarily $b \neq 0$, since $A \succeq a_N I$. The first block equation gives $-A^{-1}B_H b = (I - \lambda A^{-1})x$. Hence

$$w_b := b^\dagger W b = 1 - 2\lambda \langle x, A^{-1}x \rangle + \lambda^2 \langle x, A^{-2}x \rangle, \quad (1 - \lambda/a_N)^2 \leq w_b \leq 1. \quad (\text{KG32})$$

Both inequalities follow from the spectrum of $I - \lambda A^{-1}$ lying in $[1 - \lambda/a_N, 1]$ and $\|x\|^2 + \|b\|^2 = 1$. Thus on the unit observed vector $b/\sqrt{\theta}$, the mass is exactly w_b/θ , between $(1 - \lambda/a_N)^2/\theta$ and $1/\theta$. This proves incoming (25)–(26) with explicit relative constants.

The full spectral resolution, compressed by Λ, L , gives $S^{-1} = \lambda^{-1}bb^\dagger + R$, $0 \preceq R \preceq D_-^2 I$. Taking its maximum eigenvalue and inverting proves

$$\frac{\lambda}{\theta + \lambda D_-^2} \leq \lambda_{\min} S \leq \frac{\lambda}{\theta}. \quad (\text{KG33})$$

The graph isometry gives $\nu_i := \lambda_i(H_{\text{eff}}) \geq \lambda_i(H)$, by the minimum principle. At the actual pole, $F(-\lambda)b = 0$ and $S = F(-\lambda) + \lambda[I + B_H^\dagger A^{-1}(A - \lambda I)^{-1}B_H]$. The bracket lies between W and $(1 - \lambda/a_N)^{-1}W$. Insert b into the corresponding generalized Rayleigh quotient to obtain

$$\lambda \leq \nu_1 \leq \frac{\lambda}{1 - \lambda/a_N}, \quad \nu_2 \geq D_-^{-2} \quad (n \geq 2). \quad (\text{KG34})$$

This agrees with the stronger all-index and multiplicity theorem RM24–31. The present advance is the explicit all-activation kernel gap, not a new claim that the prior all-index eigenvalue theorem was unavailable.

Let $Y(\tau) = \Lambda e^{-\tau H} L$, $\tau \geq 0$. Its spectral expansion and KG32 show

$$\left| \text{Tr}(WY(\tau)) - e^{-\tau\lambda} \right| \leq (1 - w_b)e^{-\tau\lambda} + (\text{Tr} W - w_b)e^{-\tau/D_-^2}. \quad (\text{KG35})$$

Indeed the sum of all fast observed weighted coefficients is exactly $\text{Tr} W - w_b$: the compressed full resolution of the identity is I_B . Moreover $1 - w_b \leq 2\lambda/a_N - (\lambda/a_N)^2 \leq 2\lambda/a_N$, and $\text{Tr} W - w_b \leq (q - 1)\|W\|$, yielding incoming (29). This proves the stronger finite tail with every mode counted. The raw observed soft coefficient remains θ , while the mass-weighted coefficient is w_b ; neither is assigned value one at finite size.

For effective heat, min-max gives $\text{Tr} e^{-\tau H} \geq \text{Tr} e^{-\tau H_{\text{eff}}}$. Pairing the first eigenvalues and discarding negative fast differences gives

$$0 \leq \text{Tr} e^{-\tau H} - \text{Tr} e^{-\tau H_{\text{eff}}} \leq \frac{\lambda/a_N}{e(1 - \lambda/a_N)} + (q - 1)e^{-\tau/D_-^2}. \quad (\text{KG36})$$

For the first term, $e^{-\tau\lambda} - e^{-\tau\nu_1} \leq \tau(\nu_1 - \lambda)e^{-\tau\lambda}$, followed by KG34 and $\sup_{t \geq 0} te^{-t} = 1/e$. If desired, its exact optimized bound is $r(1 - r)^{(1-r)/r}$, $r = \lambda/a_N \in (0, 1)$: maximize $e^{-t} - e^{-t/(1-r)}$ at $t = -(1 - r)\log(1 - r)/r$. This is at most the displayed bound and keeps its uniformity in τ .

8. The complete observed critical determinant and heat

Block elimination at z and at zero gives the exact same-cutoff ratio

$$\log \frac{\det F(z)}{\det F(0)} = \log \det(I + zH^{-1}) - \log \det(I + zA^{-1}), \quad z \geq 0. \quad (\text{KG37})$$

There is only one inverse energy eigenvalue above D_-^2 ; all m kernel inverse eigenvalues are at most a_N^{-1} . Consequently

$$-m \log(1 + z/a_N) \leq \log \frac{\det F(z)}{\det F(0)} - \log(1 + z/\lambda) \leq (q-1) \log(1 + zD_-^2). \quad (\text{KG38})$$

No $\lambda < a_N$ guard is needed for KG37-38. Also $\det(S + zW)/\det S = \det(I + zH_{\text{eff}}^{-1})$, so KG27 compares this effective determinant with KG37 with error at most $\rho \log(1 + z/a_N)$. This is a cancellation in a same-cutoff ratio, with the mass determinant KG20-21 retained separately.

For the critical limit, let $\ell_k = \log(q/k)$, and use fixed real ζ, b , or bounded parameter sets with strict heat gaps bounded away from zero. Put $\alpha_k(\zeta) = e^{-2q\ell_k - \zeta q}$, $z_k(b) = e^{-2q\ell_k - bq}$, $\tau_k(b) = z_k(b)^{-1}$. These activations lie in $[0, 1]$ eventually; their exact physical domain is $\zeta \geq -2\ell_k$. The finite harmonic theorem is uniform on the larger domain, but the following ultrasmall-scale and heat assertions have the fixed/bounded parameter scope just stated.

The retained analytic endpoint theorem, HAR41-47, and the independently proved original inverse norm ACT10-19, give

$$\begin{aligned} -\log \lambda_{N,L/H} &= 2q\ell_k + \kappa_{L/H}(\zeta)q + o(q), \quad \kappa_T(\zeta) = \min(\beta_T, \zeta), \\ \beta_L &= -2 - I_\square + 2a_0, \quad a_0 = \log(4/\pi), \quad \beta_H = \beta_L - C\partial/2, \\ I_\square &= \log(\delta^2 + \gamma^2) - 3 + (\delta/\gamma) \arctan(\gamma/\delta) + (\gamma/\delta) \arctan(\delta/\gamma). \end{aligned} \quad (\text{KG39})$$

The high-endpoint monic-profile/zero-law theorem has its retained analytic status. No new native moment evaluation is asserted. At this scale, z_k/a_N and $z_k D_-^2$ are bounded above by $\exp[-2q\ell_k + O_{h,\varpi,b}(q)]$, as are the two errors in KG38 after their polynomial dimension factors. They may vanish; these are upper bounds, not asymptotic equalities to a nonzero error.

The inequality $0 \leq \log(1 + e^x) - [x]_+ \leq \log 2$, and the 1-Lipschitz property of $[x]_+$, prove, including equality at thresholds,

$$\frac{1}{q} \mathcal{R} \log \frac{\det F_N(z_k(b))}{\det F_N(0)} \rightarrow 2[(\kappa_L(\zeta) - b)_+ - (\kappa_H(\zeta) - b)_+]. \quad (\text{KG40})$$

The identical limit holds for $q^{-1} \mathcal{R} \log \det(I + z_k(b)H_{\text{eff},N}^{-1})$, by KG27. Here $\mathcal{R}$ has the unchanged signs $+, +, -, -$ at $q-1, q, 2q-1, 2q$; the two endpoints of each type have the same leading exponent, not necessarily equal finite matrices.

For heat, $\lambda/a_N = \exp[-2q\ell_k + O(q)] \rightarrow 0$, proving KG31 eventually. Also $\log \|W\| = O(q)$ by KG22, while $\tau_k/D_-^2 = \exp[2q\ell_k + bq - o(q)]$. Thus every fast term in KG35-36 tends to zero, including its mass factor. The soft factor is $e^{-\tau_k \lambda} = \exp[-\exp(q(b - \kappa_T) + o(q))]$. Therefore

$$\mathcal{R} \text{Tr}(W_N Y_N(\tau_k(b))) \rightarrow 2, \quad \mathcal{R} \text{Tr} e^{-\tau_k(b)H_{\text{eff},N}} \rightarrow 2 \quad \text{if } \kappa_H < b < \kappa_L. \quad (\text{KG41})$$

Both limits are zero on $b < \kappa_H$ and on $b > \kappa_L$: in the first region all four soft factors tend to one and the four signs cancel, and in the second all tend to zero. If $\kappa_L = \kappa_H$, the middle region is empty. At equality to an endpoint, $o(q)$ in the exponent does not fix a step value; no such value is assigned. These are the mass-weighted full observed heat and the isometric graph heat, respectively. They do not give a limit two for the raw observed heat or for $e^{-\tau M_B^\dagger M_B}$.

Finally choose $\zeta_\diamond = \beta_L - a_0$, $b_* = -2 - I_\square = \beta_L - 2a_0$. The exact source certificate AK18 proves $C_\partial > 4a_0 > 0$. Thus $\beta_H < b_* < \zeta_\diamond < \beta_L$, and KG40 gives

$$\boxed{\mathcal{R} \log \frac{\det F_N(z_k(b_*))}{\det F_N(0)} = 2a_0q + o(q) \quad \text{at } \alpha_k(\zeta_\diamond).} \quad (\text{KG42})$$

At the canonical endpoint $\alpha = 0$, the same source formula has $\kappa_T = \beta_T$, so the value is $4a_0q + o(q)$. Their difference is $-2a_0q + o(q)$, with the same regularizer at all cutoffs and both activations. KG41 holds at b_* in both cases. This verifies the incoming coefficients and signs. The gap has supplied the missing complete observed receiver; it has not supplied an initial kernel-volume coefficient or an individual complex-current phase.

9. Verification and affected results

The complete finite arguments above verify received (1), (4)–(29), (31)–(32), and (36) on their stated guards; (30) follows from KG36; the critical conclusions (2)–(3), (33)–(35) follow from KG39–42 with the precise limiting scope. The only needed corrections are the scope of the full ultrasmall scale over activation, the distinction between an inherited-metric inclusion and the coefficient frame, and the relative-base-metric meaning of subquotient mass costs. No coefficient error is present in the concrete $2a_0$, $4a_0$, or $-2a_0$ claims.

KG11 improves the guarded canonical gap ME29 and the gap receivers RM19/RM23 by using the original inverse residual directly. KG13–16 extends BR1–5 to all activations and the two growing word families. KG20–22 extends ME31–34 and BR6 to every activation; KG24 extends ME20’s memory bound. KG25–38 gives low-frequency and heat receivers for the original complete pencil, while the exact all-index pole reconstruction of RM24–31 remains valid. BR7–8’s independent compressed-action leakage is not removed or reinterpreted by these identities.

The accompanying checker distinguishes exact rational polynomial-source and geometric fixtures from numerical matrix comparisons. No fixture is called an arithmetic zero-packet or period evaluation. Source-use identities and coverage are recorded in the source-use section of `SOURCE_LEDGER.json`; every imported programme result above points to its actual pinned proof file.

Part II

Moving rational families and full filtered heat

Uniform activation, complete rational quotients, and filtered heat

21 September 2026. Independent proof of the new activation, quotient-filter and spectral claims in the received moving-family continuation, sections 1–5 and 7–8. Its original attachment has SHA-256 E9ABFCCC8CC9F7D81A9B77486987FBD7EBD3CADB890E ABCFECFB5C6FD8A9B66A. The global polynomial continuation at a vanishing resultant and the source-complex homotopy are treated by the accompanying derivation.

The canonical collision-uniform rational norm was already proved in RF1–67. The inverse-power harmonic comparison and its critical coefficient were already proved in HAR1–49. The new statements here are the harmonic comparison for every retained rational denominator, the uniform activated quotient-filter system with its complete observation feedback, and its complete low-spectrum and heat bounds. Two further consequences are proved: all of the least s filter singular values have the same harmonic scalar up to $e^{o(q)}$, uniformly over every physical activation; and the full observed heat operator is approximated in trace norm by its compression to the observed rational sector.

1. Exact source, physical coordinates, and finite inputs

Keep the original simple-quartet and period domain:

$$k \geq 9, \quad k \equiv 1 \pmod{4}, \quad q = (k+1)^2, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad S = k/2 + iy,$$

$$Q_k(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \quad E = \mathbb{C}[y]/(Q_k), \quad M[f] = [yf]. \quad (\text{MF1})$$

The convolution source remains $d\mu_k = w_h^{*k}(y) dy$. Its degree- N Gram is H_N , its remainder map J_N , its attained metric $G_0 = G_N$, and its minimum section $L_N = H_N^{-1} J_N^* G_N$, for $q-1 \leq N \leq 2q$. The original observation $\Lambda : E \rightarrow B$ is onto, with full kernel K . No conductor kernel replaces K .

Here are the hypotheses behind the added-source bound, retained from J1–J7 of the complete joint-source proof. The fixed one-factor divisor h retains the complete orders of its selected original zeros. Its divided-jet unit $U_h = j_h(2\xi/h)$ is invertible, and its physical injection is the complete $U_h^{\otimes k} \beta$. The original positive source Gram and the whole one-factor relation space give a section R_η with $JR_\eta = I$, $R_\eta^* H R_\eta = \tau_\eta T_\eta$, through degree $n_{\text{sec}} = 2 \deg h - 1$. The added source is exactly $Y_{k,\eta} = \text{im}(\mathcal{A}_h R_\eta)^{\otimes k} \beta$; the same section is fixed at every cutoff. The joint source is the sum $X_N + Y_{k,\eta}$, with its original cross Gram and the constraint on the complete tensor value. It is not an orthogonal sum of independent source minima.

Use the original simple-quartet choice $\eta = 1$. Put $n_* = \max(2q, n_{\text{sec}})$,

$$\mathbf{L}_k = \Lambda_{h,n_*}^k, \quad \mathbf{B}_k = \tau_1^k, \quad T_k = G_{k,1}.$$

The original one-factor constants satisfy $\Lambda_{h,n_*} \geq 1$, $\tau_1 > 1$. Tensoring its positive full-form inequality before minimizing, and using the actual section for the upper side,

proves

$$\mathbf{L}_k^{-1}T_k \preceq G_1 := G_N^J \preceq \mathbf{B}_k T_k, \quad G_1 \preceq G_0, \quad \log \mathbf{L}_k + \log \mathbf{B}_k = O_h(k \log^2(q+2)). \quad (\text{MF2})$$

In particular both individual logarithms have this bound, because both are nonnegative. The separate degree n_* , the fixed divisor and section, and the complete value constraint are part of this assertion. An arbitrary positive added source is not assigned MF2 without these hypotheses.

In the complete root-value coordinates,

$$T_k = \text{diag}(W_\omega), \quad W_\omega \geq a_{\min}^k > 0, \quad \sum_{\omega} W_\omega = A_w^k.$$

These are the original ordered-tuple weights. This value-coordinate expression is transported through the same physical unit as above; it is not a replacement coefficient metric.

Write the exact covariance

$$C_\alpha = G_\alpha^{-1} = (1 - \alpha)G_0^{-1} + \alpha G_1^{-1} = G_0^{-1} + \alpha \Omega_N, \quad 0 \leq \alpha \leq 1, \\ \Omega_N = F_Z H_Z^\dagger F_Z^* \succeq 0, \quad Q_\alpha = (\Lambda C_\alpha \Lambda^*)^{-1}, \quad \mathcal{L}_\alpha = C_\alpha \Lambda^* Q_\alpha, \quad P_{B,\alpha} = \mathcal{L}_\alpha \Lambda. \quad (\text{MF3})$$

Here $H_Z = \Gamma - C_X^* H_X^{-1} C_X$ and $F_Z = I - J_N H_X^{-1} C_X$ are the complete original normal-source matrices. The positive-range inverse retains every null direction. The maps in MF3 are the full attained minima; $\Lambda = \mathcal{L}_\alpha^\dagger$ and $\mathcal{L}_\alpha$ is an isometry for Q_α, G_α .

Fix $\mathcal{P} \in \mathbb{C} \setminus \mathcal{L}$, where $\mathcal{L} = \{(2a+1)\gamma - i(2b+1)\delta : a, b \in \mathbb{Z}\}$. Let D be monic of degree d , with all roots in $\mathcal{P}$, including all repetitions, and put $U_D a = [a/D]$, $\deg a < d$, $V_D = \text{ran } U_D$. The canonical result used is RF16-41 in sealed016:

$$a_{d,N} S_N I_d \preceq H_{D,N}^B(0) \preceq H_{D,N}^E(0) \preceq b_{d,N} S_N I_d, \\ S_N = u_k T_N^2, \quad T_N = \frac{1}{|Q_k(0)| \sqrt{K_{Q_k, n_0}(0, 0)}}, \quad n_0 = 2 \left\lfloor \frac{N+1-q}{2} \right\rfloor, \\ a_{d,N} = \frac{\ell_{k,N}}{u_k} \left(\frac{\mathcal{L}_N^{\text{RF}}}{2\mathcal{F}_N^{\text{RF}}(\mathcal{B}_N^{\text{RF}})^{Jd}} \right)^2, \quad b_{d,N} = (\mathcal{U}_{d,N}^{\text{RF}})^2. \quad (\text{MF4})$$

The superscript identifies exactly RF19, RF32-34 and RF39; it avoids confusing those scalar constants with the observed section $\mathcal{L}_\alpha$. All depend on the fixed compact pole set and degree, not on distances between individual poles.

For clarity, the complete finite domain carried forward is RF37:

$$q_- = (k-7)^2 \geq 16, \quad R_k = k\sqrt{\delta^2 + \gamma^2} \leq 2^{-14} q_-, \quad a_\Gamma^2 \leq \beta_q/2, \quad \beta_q = 2^{-27} q^2, \\ Jd - v \leq q_-, \quad n_- + 2 \leq 2q_-, \quad n_- = N - v + Jd - q_-, \quad \mathcal{L}_N^{\text{RF}} T_N^{\text{lb}} \geq 2\mathcal{A}_d^{\text{RF}}. \quad (\text{MF5})$$

Here a_Γ is the radius of RF6's fixed punctured contour and

$$T_N^{\text{lb}} = \frac{\sqrt{c_\sigma q/(1+16q^2)}(q - R_k)^q e^{-\pi q/2}}{|Q_k(0)|(n_0+1)10^{n_0}}, \quad c_\sigma = \Gamma(1/4)^2/(2\pi).$$

RF32 explicitly retains the entire boundary product and every nearby-degree kernel factor in $\mathcal{L}_N^{\text{RF}}$. The inequality $n_- - n_0 = \Delta + Jd - v + (N - q - n_0) \geq \Delta > 0$ shows that MF5 covers both complex kernel degrees. IVO10 imposes no additional radius restriction on its complex evaluation, whereas the separate reciprocal- H guard is explicitly present in MF5. These finite guards hold eventually when $d \log(q+2) = o(q)$, with the original fixed packet, period and nonzero conductor symbol. Uniformity over moving periods is not inferred if the first nonzero moment or source constants lose their stated bounds.

2. Uniform harmonic enclosure for the entire rational family

Put $R_P = \max_{\lambda \in \mathcal{P}} |\lambda|$, $a_* = \text{dist}(\mathcal{P}, \mathcal{L}) > 0$, $A = 2 \max(1, R_P)$, and

$$C_{\mathcal{P}} = \max(2, A/a_*).$$

For $|\omega| \leq A$, $|a(\omega)| \leq \sqrt{d}A^{d-1}\|a\|_2$ and $|D(\omega)| \geq a_*^d$. For $|\omega| > A$, each denominator factor is at least $|\omega|/2$, whereas the numerator is at most $\sqrt{d}|\omega|^{d-1}\|a\|_2$. Therefore, with every original root and weight retained,

$$|a(\omega)/D(\omega)| \leq \sqrt{d}C_{\mathcal{P}}^d\|a\|_2, \quad U_D^*T_kU_D \preceq dC_{\mathcal{P}}^{2d}A_w^kI_d. \quad (\text{MF6})$$

The full cardinal-lift estimate HAR13-22 gives

$$\text{Tr}(T_k^{-1}G_0) \leq \Gamma_N S_N, \quad \Gamma_N = \max\left(1, \frac{qR_0^2}{\delta^2 a_{\min}^k} e^{8kR_0/\delta + 2n_0R_k^2/\beta_q}\right), \quad R_0 = \sqrt{\delta^2 + \gamma^2}. \quad (\text{MF7})$$

Its guards follow from MF5: $R_k^2 \leq \beta_q/2$ and $n_0 \leq q \leq 2q - 2$. For completeness its source lift is $Q_k H_{Q_k, n_0} / [(y - \omega)Q'_k(\omega)H_{Q_k, n_0}(\omega)]$. The root product proves $|Q_k(0)/Q'_k(\omega)| \leq R_0 e^{4kR_0/\delta}$; $|y - \omega| \geq \delta$ on the real line and the full Gamma comparison give the corresponding diagonal source norm. Summing these norms divided by W_ω gives exactly the trace in MF7. The operator inequality $T_k^{-1} \preceq \text{Tr}(T_k^{-1}G_0)G_0^{-1}$ follows by congruence from a positive operator being at most its trace times the identity. It uses the entire covariance.

Invert the lower half of MF2, then apply MF7. Thus

$$G_1^{-1} \preceq \mathbf{L}_k \Gamma_N S_N G_0^{-1}, \quad G_\alpha \succeq \frac{G_0}{1 - \alpha + \alpha \mathbf{L}_k \Gamma_N S_N}. \quad (\text{MF8})$$

The same lower inequality holds for Q_α , by applying the fixed Λ, Λ^* to the covariance inequality and then inverting, or by taking the same affine minimum on the metric side.

Put

$$h_N(\alpha) = \frac{S_N}{1 - \alpha + \alpha S_N}, \quad a_{d,N} = \frac{a_{d,N}}{\max(1, \mathbf{L}_k \Gamma_N)}, \quad \bar{b}_{d,N} = b_{d,N} + \mathbf{B}_k d C_{\mathcal{P}}^{2d} A_w^k.$$

MF4 and MF8 give the lower bound. For the upper bound use both $G_\alpha \preceq G_0/(1 - \alpha)$ and $G_\alpha \preceq G_1/\alpha$, omitting a bound with zero denominator. The elementary identity

$$\min\left(\frac{bs}{1 - \alpha}, \frac{t}{\alpha}\right) \leq (b + t) \frac{s}{1 - \alpha + \alpha s}$$

is verified by considering which fraction is smaller; it extends to both endpoints. Consequently

$$\boxed{a_{d,N} h_N(\alpha) I_d \preceq H_{D,N}^B(\alpha) \preceq H_{D,N}^E(\alpha) \preceq \bar{b}_{d,N} h_N(\alpha) I_d.} \quad (\text{MF9})$$

Take

$$E_{d,N} = \max(0, -\log a_{d,N}, \log \bar{b}_{d,N}), \quad E_{k,d} = \max_N E_{d,N}.$$

MF2, MF4, MF6 and MF7 prove

$$E_{k,d} = O_{h,A,\mathcal{P}}\left(k \log^2(q + 2) + (k + Jd) \log(q + Jd + 2)\right). \quad (\text{MF10})$$

In particular it is $o(q)$ in the stated degree regime and independent of α . This is the received harmonic theorem, now for the exact joint source specified in MF2-3.

3. Critical activation, subquotients, and the complete kernel

The scalar endpoints were already evaluated in HAR41-49:

$$\log S_T = 2q\ell_k + \beta_T q + e_{T,k}, \quad \ell_k = \log(q/k), \quad \epsilon_k^{\text{end}} = \max_T |e_{T,k}| = o(q),$$

$$T = L, H, \quad \beta_L = 2\log(4/\pi) - 2 - I(\delta, \gamma), \quad \beta_H = \beta_L - \mathfrak{C}/2, \quad 1.353893 < \mathfrak{C} < 1.354751. \quad (\text{MF11})$$

The original high-endpoint profile and zero-law theorem keeps the exact source status stated there. It is not proved again by the present finite matrix bounds.

For $\alpha_k(\zeta) = e^{-2q\ell_k - \zeta q}$, the exact physical domain is $\zeta \geq -2\ell_k$, with $\alpha = 0$ represented by $\zeta = +\infty$. For $s \geq 1$,

$$\max(1, \alpha s) \leq 1 - \alpha + \alpha s \leq 2 \max(1, \alpha s).$$

Thus, once both scalar endpoints are at least one, every Gram eigenvalue obeys

$$|\log \lambda_j(H_{D,N}^X(\alpha_k(\zeta))) - 2q\ell_k - q \min(\beta_{L/H}, \zeta)| \leq E_{d,N} + \epsilon_k^{\text{end}} + \log 2, \quad X = E, B. \quad (\text{MF12})$$

This is uniform on the entire moving physical domain, not only on bounded ζ -intervals. It yields

$$\begin{aligned} \mathcal{R} \log \det H_{D,N}^X(\alpha_k(\zeta)) &= dqC_{\text{inv}}(\zeta) + O\left(d \sum_N E_{d,N} + 4d\epsilon_k^{\text{end}} + 4d \log 2\right), \\ C_{\text{inv}}(\zeta) &= 2[\min(\beta_L, \zeta) - \min(\beta_H, \zeta)]. \end{aligned} \quad (\text{MF13})$$

The finite all-activation version replaces the scalar term by $2d \log(h_L(\alpha)/h_H(\alpha))$ and has error $d \sum_N E_{d,N}$.

For any fixed full-column coefficient map $W : \mathbb{C}^r \rightarrow \mathbb{C}^d$ and onto map $A : \mathbb{C}^r \rightarrow \mathbb{C}^t$, restrict MF9 and take the full affine minimum. Their exact coefficient metric

$$Q_{\text{coef}} = (A(W^*W)^{-1}A^*)^{-1}$$

remains in

$$e^{-E_{d,N}} h_N Q_{\text{coef}} \preceq [A(W^* H_{D,N}^X W)^{-1} A^*]^{-1} \preceq e^{E_{d,N}} h_N Q_{\text{coef}}. \quad (\text{MF14})$$

Each inverse reverses order, proving the claim with the same fibre. Determinants have rank t in the error. The coefficient determinant cancels at the four cutoffs when the coefficient maps are identical there; it is not discarded if those maps change.

MF9 proves $K \cap V_D = 0$. With any fixed kernel frame I_K , form the full cross Gram $C = I_K^* G_\alpha U_D$. The two Schur complements of the Gram of $[I_K, U_D]$ prove

$$\delta_{D,N}(\alpha) = \log \frac{\det H_{D,N}^E(\alpha)}{\det H_{D,N}^B(\alpha)} = \log \frac{\det(I_K^* G_\alpha I_K)}{\det[I_K^* G_\alpha I_K - C(H_{D,N}^E(\alpha))^{-1} C^*]}. \quad (\text{MF15})$$

After orthonormalizing the two frames, only $\min(\dim K, d)$ cross singular values are nonzero. The relative eigenvalues in MF9 are at least $e^{-2E_{d,N}}$. Hence

$$0 \leq \delta_{D,N}(\alpha) \leq 2 \min(\dim K, d) E_{d,N}. \quad (\text{MF16})$$

For $\dim K = 8k - 16$, all four angle losses are $o(kq)$. At $d = \lfloor q/\log^2(q+2) \rfloor$ the displayed bound is $O(k^2 \log^2 q + Jkq/\log q) = o(kq)$, uniformly over all activations and pole multisets. The quotient retains the kernel metric; it does not evaluate its unknown determinant.

4. Exact filter quotients and their activated norm

For a monic D of degree $d \leq q$, define the fixed coefficient quotient

$$\pi_D[x] = \text{quo}_D \text{rem}_{Q_k}(Dx) \in \mathcal{P}_{<q-d}. \quad (\text{MF17})$$

On the resolvent domain, its kernel is V_D , and the low-polynomial inclusion is a right inverse: if $\deg x < q - d$, then $\deg Dx < q$ and the quotient is exactly x . Its metric is

$$\overline{G}_D(\alpha) = (\pi_D C_\alpha \pi_D^*)^{-1}.$$

The case $D = 1, d = 0$ has $\pi_D = I$ and $V_D = 0$.

Let F be monic of degree s , all roots in $\mathcal{P}$, with $d + s \leq q$. Define

$$\mathcal{I}_{D,F}[x] = [F(M)^{-1}x] \in E/V_{DF}.$$

It is well defined because $F(M)^{-1}V_D \subset V_{DF}$. The complete coefficient identity is

$$\boxed{\pi_{DF} F(M)^{-1} = \text{quo}_F \pi_D.} \quad (\text{MF18})$$

Indeed the left side divides $\text{rem}_{Q_k}(Dx)$ by DF ; dividing first by D , then by F , produces the same quotient because the remaining polynomial has degree below $d + s$. Its kernel in π_D -coordinates is $\mathcal{P}_{<s}$, and multiplication by F is a right inverse. It follows in particular that

$$\mathcal{I}_{DF,G} \mathcal{I}_{D,F} = \mathcal{I}_{D,FG} \quad (\text{MF19})$$

whenever all degrees fit. The polynomial identity $\pi_{DF} = \text{quo}_F \pi_D F(M)$ remains valid without inverses; its boundary interpretation is handled by the separate global-frame proof.

For the canonical endpoint, divide the actual minimum representative:

$$F(M)^{-1} = \mathbf{B}_0 + U_F T_{F,0}, \quad \mathbf{B}_0 = J_N \text{quo}_F L_N, \quad T_{F,0} = \text{rem}_F L_N,$$

$$\|\mathbf{B}_0\|_{G_0} \leq b_0 := \sqrt{u_k/\ell_{k,N}} \mathcal{D}(N, R_P)^s. \quad (\text{MF20})$$

Here $\mathcal{D}$ is RF14's complete polynomial divided-difference bound, or the valid larger constant (21) in the received manuscript. Only one full source comparison is used. The remainder lies in $V_F \subset V_{DF}$, so it is killed by the target quotient. Taking the minimum source lift in E/V_D proves $\|\mathcal{I}_{D,F}\|_{\alpha=0} \leq b_0$.

In the complete value metric T_k , $F(M)^{-1}$ is diagonal with entries $F(\omega)^{-1}$, whose modulus is at most a_*^{-s} . MF2 gives

$$\|F(M)^{-1}\|_{G_1} \leq b_1 := \sqrt{\mathbf{L}_k \mathbf{B}_k} a_*^{-s}. \quad (\text{MF21})$$

Taking the full quotient minima proves the same bound on $\mathcal{I}_{D,F}$ at $\alpha = 1$.

If A denotes its fixed matrix quo_F , a norm bound between two positive metrics is equivalent to

$$A \overline{G}_D^{-1} A^* \preceq b^2 \overline{G}_{DF}^{-1}.$$

This follows by conjugation to Euclidean coordinates and the equality of the operator norms of a matrix and its adjoint. Both quotient covariances are linear images of C_α . Taking the convex combination of the two endpoint inequalities therefore proves

$$\boxed{\|\mathcal{I}_{D,F}\|_\alpha \leq B_{s,N} := \max(b_0, b_1), \quad 0 \leq \alpha \leq 1,}$$

$$\log^+ B_{s,N} = O_{h,\mathcal{P}}(k \log^2(q+2) + s \log(q+s+2)). \quad (\text{MF22})$$

This proof works even when the numerator metrics are poorly conditioned. If the target has dimension zero, the map and norm are zero; the same bound holds.

Let $\bar{\Lambda}_D : E/V_D \rightarrow B/\Lambda V_D$ and $\bar{\mathcal{L}}_D$ be the actual quotient observation and its minimum section, in the induced metrics. They are respectively a contraction and an isometry. For $\mathcal{B}_{D,F} = \bar{\Lambda}_{DF} \mathcal{I}_{D,F} \bar{\mathcal{L}}_D$, insert $\bar{\mathcal{L}}_{DF} \bar{\Lambda}_{DF} = I - P_{\ker \bar{\Lambda}_{DF}}$ into MF19. This gives exactly

$$\mathcal{B}_{D,FG} - \mathcal{B}_{DF,G} \mathcal{B}_{D,F} = \bar{\Lambda}_{DFG} \mathcal{I}_{DF,G} P_{\ker \bar{\Lambda}_{DF}} \mathcal{I}_{D,F} \bar{\mathcal{L}}_D. \quad (\text{MF23})$$

The kernel equals the image of K in E/V_{DF} , so its dimension is at most $\dim K$, with equality on the observed rational domain. Thus the feedback rank is at most $\dim K$, and its norm is at most $B_{\deg F, N} B_{\deg G, N}$. Its complete projection, all metrics, and all original kernel directions remain.

5. A full activated inverse decomposition

The preceding quotient argument has an exact operator realization extending ACT7-9. Define

$$\mathcal{J}_\alpha(x_0, x_1) = \sqrt{1-\alpha} x_0 + \sqrt{\alpha} x_1 : E_{G_0} \oplus E_{G_1} \rightarrow E_{G_\alpha}.$$

Its actual adjoint is

$$\mathcal{J}_\alpha^\dagger x = (\sqrt{1-\alpha} G_0^{-1} G_\alpha x, \sqrt{\alpha} G_1^{-1} G_\alpha x).$$

MF3 proves $\mathcal{J}_\alpha \mathcal{J}_\alpha^\dagger = I$. Consequently every fixed coefficient operator A satisfies

$$A = \mathcal{J}_\alpha \text{diag}(A, A) \mathcal{J}_\alpha^\dagger, \quad \|A\|_{G_\alpha} \leq \max(\|A\|_{G_0}, \|A\|_{G_1}). \quad (\text{MF24})$$

The identity retains the factor order $AC_\alpha G_\alpha = A$.

Put $X = F(M)^{-1}$ and use the complete canonical decomposition MF20. Then

$$X = \mathbf{B}_\alpha + (1-\alpha) U_F T_{F,0} G_0^{-1} G_\alpha,$$

$$\mathbf{B}_\alpha = (1-\alpha) \mathbf{B}_0 G_0^{-1} G_\alpha + \alpha X G_1^{-1} G_\alpha, \quad \|\mathbf{B}_\alpha\|_{G_\alpha} \leq B_{s,N}. \quad (\text{MF25})$$

The bound follows by inserting $\text{diag}(\mathbf{B}_0, X)$ into the same coisometry. The remainder has image in V_F and rank at most s . It vanishes at $\alpha = 1$. Thus no claim of an activation-uniform separated rank- s cluster follows merely from its rank.

Applying $I - P_{V_F, \alpha}$ proves

$$\|(I - P_{V_F, \alpha})X\| \leq B_{s,N}, \quad \sigma_{s+1}(X) \leq B_{s,N} \quad (s < q). \quad (\text{MF26})$$

The second statement follows from the singular-value minimum principle on the kernel of the rank- s remainder. It requires no norm bound for that remainder.

6. The entire low spectrum and the full observed heat

Let $H_F = X^{-\dagger} X^{-1} = F(M)^\dagger F(M)$, and let $P_{\leq \lambda}$ be its full spectral projection. All adjoints and projections in this section use the actual G_α . For x in that range, $\|X^{-1}x\|^2 = \langle H_F x, x \rangle \leq \lambda \|x\|^2$. MF26 therefore proves

$$\|(I - P_{V_F, \alpha})P_{\leq \lambda}\| \leq B_{s,N} \sqrt{\lambda}, \quad \text{rank } P_{\leq \lambda} \leq s \quad (0 < \lambda < B_{s,N}^{-2}). \quad (\text{MF27})$$

The strict spectral threshold is essential for the rank statement with a closed spectral projection.

Write

$$\ell_{F,N,\alpha} = \lambda_{\min}[(H_{F,N}^E)^{-1/2} H_{F,N}^B (H_{F,N}^E)^{-1/2}] \geq e^{-2E_{k,s}}.$$

For $\varepsilon = B_{s,N}\sqrt{\lambda} < 1$, the exact two-subspace estimate gives every unit $x \in \text{ran } P_{\leq \lambda}$

$$\|P_{B,\alpha}x\|^2 \geq \left(\sqrt{\ell}\sqrt{1-\varepsilon^2} - \sqrt{1-\ell\varepsilon}\right)_+^2. \quad (\text{MF28})$$

To prove it, decompose x into its V_F and orthogonal components. For unit $z \in K$, $|\langle z, x \rangle| \leq a\sqrt{1-t^2} + \sqrt{1-a^2}t$, where $a = \|P_{V_F}z\| \leq \sqrt{1-\ell}$, $t \leq \varepsilon$. For $\varepsilon < \sqrt{\ell}$, the expression increases on both intervals; substituting their endpoints and subtracting its square from one gives MF28. For larger ε the positive-part bound is zero. Thus, at $\lambda = e^{-cq}$, $c > 0$, the observed fraction is at least $\ell(1-o(1))$, uniformly in activation and pole family. The explicit ratio $\varepsilon/\sqrt{\ell} \leq B_{s,N}e^{-cq/2+E_{k,s}} \rightarrow 0$ proves this uniformity. The low space may be zero.

Let P_R be any original orthogonal projection annihilating V_F . For left singular vectors a_j of X , $\|P_R a_j\| \leq \min(1, B_{s,N}/\sigma_j)$. Since a_j are precisely the heat eigenvectors,

$$\text{Tr}(P_R e^{-\tau H_F}) = \sum_j e^{-\tau/\sigma_j^2} \|P_R a_j\|^2.$$

For the first s terms use $\sup_{x \geq 0} x e^{-x} = 1/e$. For the rest use $\sigma_j \leq B_{s,N}$. Therefore

$$0 \leq \text{Tr}(P_R e^{-\tau H_F}) \leq \varepsilon_{s,N}(\tau) := s \min(1, B_{s,N}^2/(e\tau)) + (q-s)e^{-\tau/B_{s,N}^2}. \quad (\text{MF29})$$

The last term is absent if $s = q$; at $\tau = 0$ use the limiting bound q . For $\tau = e^{cq}$, this is at most $e^{-cq+o(q)}$, because $\log q = o(q)$ and $\log^+ B_{s,N} = o(q)$.

If $F \mid D$, then $V_F \subset V_D$: multiply its numerator by D/F , whose degree is $d-s$. Set $P_R = P_{(K+V_D)^\perp}$. The actual orthogonal decomposition is

$$E = K \oplus P_{B,\alpha}V_D \oplus (K + V_D)^\perp.$$

This follows because $K + P_{B,\alpha}V_D = K + V_D$, the first two summands are orthogonal, and $K \cap V_D = 0$. The complete observed heat is $\mathcal{H}_B(\tau) = \Lambda e^{-\tau H_F} \mathcal{L}_\alpha$. Let Π_D be the Q_α -orthogonal projection onto ΛV_D . Isometry of $\mathcal{L}_\alpha$ identifies it with $P_{P_{B,\alpha}V_D}$. Thus

$$0 \leq \text{Tr}_B \mathcal{H}_B - \text{Tr}_B(\Pi_D \mathcal{H}_B \Pi_D) = \text{Tr}_E(P_R e^{-\tau H_F}) \leq \varepsilon_{s,N}(\tau). \quad (\text{MF30})$$

This proves the received complete trace localization, without replacing the full filtered heat by the heat of a compressed action.

There is also a full operator estimate. Since $0 \preceq \mathcal{H}_B \preceq I_B$, its blocks under $\Pi_D \oplus (I - \Pi_D)$ satisfy $\text{Tr } D \leq \varepsilon_{s,N}$ and $\text{Tr } A \leq d$. Factoring the positive operator as a square and applying $\|UV^*\|_1 \leq \|U\|_{\text{HS}}\|V\|_{\text{HS}}$ gives $\|C\|_1 \leq \sqrt{\text{Tr } A \text{Tr } D}$. Hence

$$\|\mathcal{H}_B - \Pi_D \mathcal{H}_B \Pi_D\|_1 \leq \varepsilon_{s,N}(\tau) + 2\sqrt{d\varepsilon_{s,N}(\tau)}. \quad (\text{MF31})$$

At $\tau = e^{cq}$, its right side is $e^{-cq/2+o(q)}$. This retains the off-diagonal observed coupling, not only the scalar trace.

7. Every least filter singular value has the harmonic scale

The preceding results only need a small-rank bounded decomposition. The rational norm additionally determines the complete least- s scale, uniformly at all activations.

Fix F of degree $1 \leq s \leq q$, and retain its canonical coefficient constants from RF8,13,40,44,47:

$$P_s = \sqrt{M_\sigma} \left(\sum_{j=0}^{s-1} [2(s+1)]^{2j} \right)^{1/2}, \quad \rho_{s,N} = \frac{C_s E_\sigma(N, a_\Gamma)}{\sqrt{\ell_{k,N}}},$$

so that $\| [a] \|_{G_0} \leq \sqrt{u_k} P_s \|a\|_2$ and $\|T_{F,0}\| \leq \rho_{s,N}$. Both apply to every original numerator, including repeated poles. MF4 and MF20 give

$$\|X\|_{G_0} \leq b_0 + \sqrt{b_{s,N} S_N} \rho_{s,N}.$$

Define the positive finite constants

$$c_{s,N} = \max \left[\left(b_0 / \sqrt{S_N} + \sqrt{b_{s,N}} \rho_{s,N} \right)^2, b_1^2 / S_N \right], \quad C_{s,N}^X = c_{s,N} + b_1^2. \quad (\text{MF32})$$

By MF24, $\|X\|_{G_\alpha}^2 \leq c_{s,N} S_N$. For $\alpha > 0$, the separate joint comparison gives

$$\|Xx\|_{G_\alpha}^2 \leq \alpha^{-1} \|Xx\|_{G_1}^2 \leq \alpha^{-1} b_1^2 \|x\|_{G_1}^2 \leq \alpha^{-1} b_1^2 \|x\|_{G_\alpha}^2.$$

Applying the same scalar harmonic inequality as in MF9 proves

$$\boxed{\|F(M)^{-1}\|_{G_\alpha}^2 \leq C_{s,N}^X h_N(\alpha)}. \quad (\text{MF33})$$

Conversely $F(M)U_F a = [a]$ exactly. Since $G_\alpha \preceq G_0$, the actual restriction to V_F obeys

$$\frac{\|F(M)U_F a\|_{G_\alpha}^2}{\|U_F a\|_{G_\alpha}^2} \leq \frac{u_k P_s^2 e^{E_{s,N}}}{h_N(\alpha)}.$$

The singular-value minimum principle on this s -dimensional subspace, together with MF33, proves simultaneously for every $1 \leq j \leq s$

$$\boxed{\frac{1}{C_{s,N}^X h_N(\alpha)} \leq s_{q-s+j,N}(F(M); G_\alpha)^2 \leq \frac{u_k P_s^2 e^{E_{s,N}}}{h_N(\alpha)}}. \quad (\text{MF34})$$

This finite statement needs MF5 for the rational enclosure but does not require the canonical rank/dominance guards RF49.

Put

$$J_{s,N} = \max\{0, \log C_{s,N}^X, \log(u_k P_s^2) + E_{s,N}\}.$$

The unchanged u_k has $|\log u_k| = O_h(k + \log q)$, as proved from the full constant-polynomial source comparison and its ratio in HAR40. Thus $|\log \ell_{k,N}| = O_h(k + \log q)$ as well. Since $S_N \geq 1$ eventually, every displayed factor in MF32 has logarithmic cost

$$J_{s,N} = O_{h,A,\mathcal{P}} \left(k \log^2(q+2) + (k+Js) \log(q+Js+2) \right) = o(q). \quad (\text{MF35})$$

The finite formula MF32 is valid before $S_N \geq 1$; that eventual fact is used only for its asymptotic logarithmic cost. MF34 now gives the uniform least-singular law

$$|\log s_{q-s+j,N}(F(M); G_\alpha)^2 + \log h_N(\alpha)| \leq J_{s,N}. \quad (\text{MF36})$$

At $\alpha = 1$, $h_N(1) = 1$. These directions then have only $e^{o(q)}$ variation, and no separated exponentially slow cluster is asserted. Whenever $\log h_N(\alpha) \geq cq$ for a fixed $c > 0$,

MF26 and MF36 prove a separated cluster of exactly s eigenvalues, because all the others are at least $B_{s,N}^{-2} = e^{-o(q)}$.

Combining MF12's scalar denominator bound with MF36 yields, uniformly over every physical ζ ,

$$|\log s_{q-s+j,N}(F(M); G_{\alpha_k(\zeta)})^2 + 2q\ell_k + q \min(\beta_{L/H}, \zeta)| \leq J_{s,N} + \epsilon_k^{\text{end}} + \log 2. \quad (\text{MF37})$$

Consequently

$$\mathcal{R} \sum_{j=q-s+1}^q \log s_{j,N}(F(M); G_{\alpha_k(\zeta)})^2 = -sqC_{\text{inv}}(\zeta) + o(sq) \quad (\text{MF38})$$

uniformly on the whole physical activation range. Its explicit error is $s \sum_N J_{s,N} + 4s\epsilon_k^{\text{end}} + 4s \log 2$.

For the same filter at all four cutoffs, $\det(F(M)^\dagger F(M)) = |\det F(M)|^2$ is independent of metric and cutoff. Thus the remaining $q-s$ squared singular values have the opposite four-cutoff logarithmic return $+sqC_{\text{inv}}(\zeta) + o(sq)$. This exact compensation uses the original determinant, not separately adjusted errors.

8. Critical late heat of the complete polynomial family

Take a common time

$$\tau_k(b) = \exp(2q\ell_k + bq),$$

where b is fixed, or varies in a bounded set with every stated strict gap bounded below. For fixed physical ζ , or uniformly over moving ζ with those gaps, MF37 gives

$$\tau_k(b) s_{q-s+j,N}(F(M); G_{\alpha_k(\zeta)})^2 = \exp\{q[b - \min(\beta_{L/H}, \zeta)] + o(q)\}.$$

The other $q-s$ heat terms are bounded by $(q-s)e^{-\tau_k(b)/B_{s,N}^2}$. Therefore, away from the exact threshold equalities,

$$\frac{1}{s} \mathcal{R} \text{Tr} e^{-\tau_k(b)F(M)^\dagger F(M)} \longrightarrow 2 [\mathbf{1}_{\{b < \min(\beta_L, \zeta)\}} - \mathbf{1}_{\{b < \min(\beta_H, \zeta)\}}]. \quad (\text{MF39})$$

This is the full activated heat, not the heat of a compressed observation.

For the retained native time $\tau_k^* = \tau_k(b_*)$, where

$$b_* = -2 - I(\delta, \gamma),$$

the exact endpoint relations give $\beta_H < b_* < \beta_L$. Indeed $\beta_L - b_* = 2\psi(0) > 0$, and the retained strict bounds $\psi(1) < 1/8$, $J(1) < -1.163427$ give $b_* - \beta_H = -2[1 + \psi(1) + J(1)] > 0$. Thus

$$\frac{1}{s} \mathcal{R} \text{Tr} e^{-\tau_k^* F(M)^\dagger F(M)} \longrightarrow \begin{cases} 2, & \zeta > b_*, \\ 0, & \zeta < b_*. \end{cases} \quad (\text{MF40})$$

The equality $\zeta = b_*$ is not assigned a value without finer original asymptotics. The convergence is uniform when the indicated threshold gaps are bounded below. It is not asserted uniformly through that threshold.

When $\zeta > b_*$ with a fixed positive margin, choose an exponentially small spectral threshold that includes every low-cutoff slow eigenvalue. MF27–28 gives each such vector an observed fraction at least $e^{-o(q)}$. The high-cutoff heat is at most $qe^{-\exp(cq)}$ for some $c > 0$. Therefore the original measured return is eventually positive, and

$$\frac{1}{q} \log \left[\frac{1}{s} \mathcal{R} \text{Tr}_B (\Lambda e^{-\tau_k^* F(M)^\dagger F(M)} \mathcal{L}_\alpha) \right] \longrightarrow 0. \quad (\text{MF41})$$

For $\zeta < b_*$, the absolute measured return tends to zero at the same double-exponential bound furnished by the full heat; no sign is imposed. At every activation, independently of these critical heat limits, MF30–31 retain their uniform finite localization bounds.

9. Constants in the received canonical reconstruction

The received sections 2–3 reproduce the canonical result already proved in RF1–41. Their alternative displayed constants are compatible with that result, with the following precise domains.

The contour formula requires the integrand to be holomorphic on a neighbourhood of the closed contour domain, not merely in its interior if boundary values are undefined. The actual $a/(Q_k H_{Q_k, n_0})$ meets this requirement under MF5. The fixed punctured contour in RF6 is sufficient; a polygonal perturbation smaller than its positive separations also works.

The stated larger Gamma estimate (20) follows from RF13:

$$M_\sigma K_{\sigma, L}(z, z) \leq e^{2+\pi R/2} (L+1)^2 (2L+1)^{R+1/2} \leq e^2 2^{2+2R} e^{\pi R} (L+1)^{3+2R}. \quad (\text{MF42})$$

The complete common-denominator remainder bound (35) follows from the coefficient one-norm version of the contour formula, with $R_C = 1 + R_B$: its contour length contributes R_C , each divided-polynomial coefficient is bounded using $(1 + R_B)^L = R_C^L$, and summing its L coefficient rows gives LR_C^{2L} . The numerator recovery constant (32) follows by applying Cauchy’s estimate to the reversed denominator $\prod_{\nu=1}^L (1 - \xi_\nu z)^{-1}$, not to the unreversed polynomial $1/B_D(z)$. This reversed expression is the precise meaning required by that sentence. On $|z| = 1/[2 \max(1, R_B)]$, its modulus is at most 2^L . The full factorial Laurent convolution and polynomial-part recovery then give the displayed factor. Its implicit constant depends on $\mathcal{P}$ as well as the fixed conductor.

The smaller base 6^{n_0} in received (41) is valid for evaluation at zero. On the interval $[q, 2q]$, the orthonormal Legendre basis has values $\sqrt{(2j+1)/q} P_j(-3)$ at zero. Rodrigues’ finite expansion makes $P_j(3) > 0$; its recurrence gives $P_{j+1}(3) \leq 6P_j(3)$, hence $|P_j(-3)| \leq 6^j$. Thus

$$|H(0)|^2 \leq \frac{(n_0 + 1)^2 6^{2n_0}}{q} \int_q^{2q} |H(y)|^2 dy.$$

Together with $|Q_k| \geq (q/2)^q$ on this interval when $R_k \leq q/2$, the original Gamma lower envelope proves the stated $\underline{T}_N$. This refinement is unnecessary for MF5’s already proved T_N^{lb} , but it is not an error.

Finally the received conductor bound $A_\Sigma e^{Zb_N}$ is valid by the finite translation exponential and the original derivative norm. On $N \leq 2q$, its logarithm is $O_A(\sqrt{q} \log q) = O_A(k \log q)$, which fits the claimed error. The sharper polynomial-in- N bound of IVO4/RF39 remains valid as well. None of these constants authorizes a uniform bound as the original nonzero conductor moment approaches zero.

10. Exact finite verification and scope

The companion checker uses exact rational matrices for a complete polynomial source, an actual added section with its cross Gram, harmonic covariance interpolation, polynomial quotients, and complex observations. It also checks the full heat localization, harmonic least-singular bounds and trace-norm strengthening on finite numerical fixtures with their original metrics. Its executed receipt ACTIVATED_RATIONAL_CHECKS.json records 159 exact checks and 338 floating-point spectral tests. Neither type is a native xi-moment computation or a new asymptotic certificate. ACTIVATED_RATIONAL_SOURCE_USE_LEDGER.md records the exact source versions, actual reading, human citations and inherited profile-theorem status.

The new proof preserves the original added-source construction and its fixed-packet hypotheses. It establishes all-activation norm and localization bounds without assuming that the critical slow cluster survives at full activation. The strengthened harmonic singular law describes that change explicitly. The global frame at the resultant and the corrected source-complex homotopy remain with the accompanying independent derivation.

Part III

Arithmetic collisions and exact observation complexes

Polynomial sectors at arithmetic collisions and the exact observation complex

This derivation receives equations (55)–(61) and (79)–(81) of the supplied moving-family continuation. It proves the extension at every primary multiplicity and repairs the section used in its chain homotopy. The earlier quotient on the resolvent locus is RF65–67, sealed 016. The proofs below are finite algebra and metric calculations, written in full. They introduce no new analytic estimate at a pole of the arithmetic resolvent.

1. The complete polynomial frame

For an integer $t \geq 0$, write $\mathcal{P}_{<t}$ for complex polynomials of degree below t , including the zero space when $t=0$. Let Q be any monic polynomial of degree q , retaining all root multiplicities. Identify $E = \mathbb{C}[y]/(Q)$ with its unique representatives in $\mathcal{P}_{<q}$. Let M denote multiplication by y followed by remainder modulo Q . For a monic D of degree $1 \leq d \leq q$, define

$$\pi_D x = \text{quo}_D \text{rem}_Q(Dx), \quad \iota : \mathcal{P}_{<q-d} \hookrightarrow E. \quad (\text{PC1})$$

Here quo is the entire quotient in monic Euclidean division; no term is discarded before taking the indicated remainder. Both maps use the original coefficient bases. Since $\deg(Dx) < q$ for x in the low polynomial space, $\pi_D \iota = I$. Consequently

$$E = V_D^{\text{flat}} \oplus \iota \mathcal{P}_{<q-d}, \quad V_D^{\text{flat}} = \ker \pi_D, \quad \dim V_D^{\text{flat}} = d. \quad (\text{PC2})$$

Define $F_D : \mathbb{C}^d \rightarrow E$ by

$$F_D e_j = y^{q-d+j} - \iota \pi_D(y^{q-d+j}), \quad 0 \leq j < d. \quad (\text{PC3})$$

Its last d coefficient rows form the identity matrix, so it is injective and its image is precisely PC2's kernel. Monic division consists only of coefficient additions and multiplications. Thus the entries of F_D are polynomial in the coefficients of D and Q , with no inverse resultant. This proves an everywhere defined frame, even on the discriminant of Q .

Let $j : \mathcal{P}_{<d} \hookrightarrow E$ denote the numerator inclusion, and let the d -by- d coefficient matrix B_D have columns

$$B_D e_j = \text{rem}_D \text{rem}_Q(Dy^{q-d+j}).$$

Subtracting D times the quotient in PC3 proves the typed identity

$$D(M)F_D = jB_D. \quad (\text{PC4})$$

The codomain basis $[j, D\iota]$ is triangular with determinant one. The domain basis $[F_D, \iota]$ has determinant $(-1)^{d(q-d)}$, because its high block is the identity and the displayed order moves d columns across $q-d$ columns. In these bases $D(M)$ is $\text{diag}(B_D, I)$. Therefore

$$\det B_D = (-1)^{d(q-d)} \det D(M) = (-1)^{d(q-d)} \text{Res}(Q, D). \quad (\text{PC5})$$

For completeness, the last equality has the convention $\text{Res}(Q, D) = \prod_{Q(\rho)=0} D(\rho)$, roots counted with their full multiplicities. On each primary block $\mathbb{C}[y]/((y - \rho)^m)$, the basis $1, y - \rho, \dots, (y - \rho)^{m-1}$ makes $D(M)$ triangular with diagonal $D(\rho)$. The Chinese remainder isomorphism follows from Bezout for the pairwise coprime primary factors. Multiplying these determinants gives exactly PC5; repeated roots require no limiting argument.

If the resultant is nonzero, $U_D = D(M)^{-1}J$ is the original rational numerator map, and PC4 yields

$$U_D = F_D B_D^{-1}, \quad \text{im } U_D = V_D^{\text{flat}}. \quad (\text{PC6})$$

2. The full primary defect and an explicit lift

Let $g = \gcd(Q, D)$ be monic, $e = \deg g$, $Q = gQ_1$, and $D = gD_1$. The coprime polynomials Q_1, D_1 have Bezout coefficients; choose T with $TD_1 \equiv 1 \pmod{Q_1}$ if Q_1 has positive degree. Then

$$0 \longrightarrow (Q/g)\mathcal{P}_{<e} \longrightarrow V_D^{\text{flat}} \xrightarrow{D(M)} J(g\mathcal{P}_{<d-e}) \longrightarrow 0 \quad (\text{PC7})$$

is exact. Indeed, $Q \mid Dx$ is equivalent to $Q_1 \mid x$, giving the left term, with exactly e independent columns. The image ideal of multiplication by D in E is the ideal generated by g : Bezout gives inclusion in both directions. A representative of degree below d belongs to this ideal exactly when it is divisible by g , because g divides Q . This proves the image in PC7. More explicitly, the lift of gp , $\deg p < d - e$, is

$$x = \text{rem}_{Q_1}(Tp), \quad \text{rem}_Q(Dx) = gp. \quad (\text{PC8})$$

It belongs to V_D by PC1. If Q_1 is constant then necessarily $e=q=d$ and the target is zero, so no T is needed. PC7 proves $\text{rank } B_D = d - e$, and identifies the entire missing multiplication image.

For example, if $D = (y - \rho)^d$ and $(y - \rho)^m$ is the full primary factor of Q , then $e = \min(d, m)$. For $d \leq m$, the entire continued sector is

$$V_D^{\text{flat}} = (Q/D)\mathcal{P}_{<d} = \ker D(M). \quad (\text{PC9})$$

For $d > m$, PC7 retains both the m -dimensional primary kernel and the nonzero image $(y - \rho)^m \mathcal{P}_{<d-m}$. In the single moving-pole family $D_z = (y - z)^d$, PC5 gives

$$\det B_{D_z} = (-1)^d Q(z)^d. \quad (\text{PC10})$$

Thus at a root ρ of multiplicity m the determinant has order md although its rank defect is $\min(d, m)$. These are two separately evaluated features of the same exact map.

3. Original metrics and the resultant factor

Fix an original cutoff and activation. Let G be the full attained positive metric on E . Let $\Lambda : E \rightarrow B$ be the original surjection, with full attained target metric

$$Q_B = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad L = G^{-1} \Lambda^* Q_B, \quad \Lambda L = I. \quad (\text{PC11})$$

All adjoints marked $*$ are coefficient conjugate transposes. The associated Hilbert-space adjoints use these stated metrics. In particular Λ is a coisometry and its norm is one. Define

$$H = F_D^* G F_D > 0, \quad A = \Lambda F_D, \quad J = A^* Q_B A \succeq 0. \quad (\text{PC12})$$

On the resolvent locus PC6 gives, exactly,

$$H_D^E = B_D^{-*} H B_D^{-1}, \quad H_D^B = B_D^{-*} J B_D^{-1}. \quad (\text{PC13})$$

If A has full column rank, taking determinants gives

$$\log \det H_D^E = \log \det H - 2 \log |\text{Res}(Q, D)|, \quad \log \det H_D^B = \log \det J - 2 \log |\text{Res}(Q, D)|. \quad (\text{PC14})$$

The full source/observation ratio is $\det J / \det H$. It extends continuously across the resultant divisor wherever G and Q_B remain positive and A retains rank d. The positive matrix $H^{-1/2} J H^{-1/2}$ has the same spectrum as the original relative Gram matrix by PC13: the coefficient map B_D^{-*} is an isometry between the respective pulled-back source forms. These spectra are the exact observed fractions. No bound uniform across a zero observed fraction follows just from PC14.

For fixed D and Q at the four original cutoffs, the resultant term is the same and its signed return is zero. If D itself differs between cutoffs, its four resultant terms must instead be retained. If A drops rank, $\det J$ is zero and a finite logarithmic ratio cannot be asserted. The maps and the retained intersection below still exist.

4. Corrected chain section in the observable case

Suppose first that $\text{rank } A = d$, and put $V = \text{im } F_D$ and $W = \Lambda V$. Define

$$h = F_D J^{-1} A^* Q_B : B \rightarrow E. \quad (\text{PC15})$$

Then $h \Lambda F_D = F_D$, so $h \Lambda$ is a projection onto V. Also Λh is the Q_B -orthogonal projection onto W: it is idempotent, has image W, and is selfadjoint in Q_B directly from its formula. Let $\rho : B \rightarrow B/W$ be the quotient map in any fixed coordinates. Equip both quotients with their full minimum metrics and define their unique minimum sections

$$L_\pi = G^{-1} \pi_D^* (\pi_D G^{-1} \pi_D^*)^{-1}, \quad L_\rho = Q_B^{-1} \rho^* (\rho Q_B^{-1} \rho^*)^{-1}. \quad (\text{PC16})$$

Empty quotients use their unique zero maps. Set

$$\bar{\Lambda} \pi_D = \rho \Lambda, \quad j_E = (I - h \Lambda) L_\pi, \quad j_B = L_\rho. \quad (\text{PC17})$$

The induced map $\bar{\Lambda}$ is well defined since $\rho \Lambda$ annihilates V. We have $\pi_D j_E = I$, $\rho j_B = I$, and

$$\Lambda j_E = (I - \Lambda h) \Lambda L_\pi = L_\rho \rho \Lambda L_\pi = j_B \bar{\Lambda}. \quad (\text{PC18})$$

For each x, $x - L_\pi \pi_D x$ belongs to V and is fixed by $h \Lambda$. Hence

$$j_E \pi_D = I - h \Lambda, \quad j_B \rho = I - \Lambda h. \quad (\text{PC19})$$

Writing the two-term differential as $d = \Lambda$ from degree zero to degree one, these are precisely $I - j p = d h + h d$. Further $p h = 0$, $h j = 0$, and $h^2 = 0$ by its degree. This is a strong deformation retract of the original complex onto its quotient by the acyclic subcomplex $[V \rightarrow W]$.

The uncorrected minimum section L_π alone generally fails PC18-19. Its projection is the G-orthogonal projection onto V-perpendicular, whereas $h \Lambda$ need not be orthogonal. PC17 is the required exact adjustment, and is also valid at an arithmetic collision with full observation rank.

Let $Q_\pi = (\pi_D G^{-1} \pi_D^*)^{-1}$. Because L_π has image orthogonal to V and $h \Lambda L_\pi$ has image in V, the adjusted source metric is

$$j_E^* G j_E = Q_\pi + (h \Lambda L_\pi)^* G (h \Lambda L_\pi). \quad (\text{PC20})$$

In the original quotient metric its determinant change lies between zero and

$$\min(d, q - d) \log(1 + \|h\|^2). \quad (\text{PC21})$$

To prove this, factor PC20 by the stated Q_π isometry. The correction is R^*R , rank at most $\min(d, q - d)$ and norm at most $\|h\|^2$ since L_π is an isometry and Λ a coisometry. The logarithm is the sum of $\log(1 + s_j(R)^2)$. If the moving-family comparison gives $J \succeq e^{-2E_{k,d,N}} H$, then the singular values of $\Lambda|_V$ are at least e^{-E} ; h is their inverse on W and zero on its orthogonal complement. Thus

$$\|h\| \leq e^{E_{k,d,N}}, \quad 0 \leq \log \frac{\det(j_E^* G j_E)}{\det Q_\pi} \leq \min(d, q - d) \log(1 + e^{2E_{k,d,N}}). \quad (\text{PC22})$$

In particular, for $d=O(k)$, q of the retained quadratic order, and $E = O(k \log^2(q + 2) + (k + Jd) \log(q + Jd + 2))$ with fixed conductor size J , summing all four errors is $O(k^2 \log^2 q) = o(kq)$. This proves that repairing the chain section preserves the leading programme allocation in that range; it does not assign the initial allocation a value.

5. Exact reduction when observation rank is lost

There is still a complete reduction when $Z = V \cap K \neq 0$, $K = \ker \Lambda$. Split $V = Z \oplus V_0$ orthogonally in G and write $W = \Lambda V$. The restriction $\Lambda|_{V_0} : V_0 \rightarrow W$ is bijective. Let h be its inverse after the Q_B -orthogonal projection onto W . Equivalently, put

$$T = Q_B^{1/2} A H^{-1/2}, \quad h = F_D H^{-1/2} T^+ Q_B^{1/2}, \quad (\text{PC23})$$

where T^+ is the Moore-Penrose inverse: in an orthonormal singular expansion $T = \sum_{j=1}^r \sigma_j u_j v_j^*$, it is $\sum_{j=1}^r \sigma_j^{-1} v_j u_j^*$. This formula proves, without suppressing a zero singular value, that $h\Lambda$ is the identity on V_0 and zero on Z , and Λh is the orthogonal projection onto W . If $r=0$ take $h=0$.

Keep L_π, L_ρ as in PC16 and define

$$j_0 = (I - h\Lambda)L_\pi, \quad z(x) = (I - h\Lambda)(I - L_\pi \pi_D)x \in Z. \quad (\text{PC24})$$

The reduced complex, projection, and section are

$$[Z \oplus \mathcal{P}_{<q-d} \xrightarrow{(z,c) \mapsto \bar{\Lambda}c} B/W],$$

$$p_E x = (z(x), \pi_D x), \quad p_B = \rho, \quad j_E(z, c) = z + j_0 c, \quad j_B = L_\rho. \quad (\text{PC25})$$

To check $z(x)$ belongs to Z , the inner factor lies in V , and $I - h\Lambda$ is its projection onto Z along V_0 . Next $\pi_D j_0 = I$, $z(j_0 c) = 0$ because $(I - h\Lambda)h = 0$, and $z(z) = z$. Thus $p_j = I$. A direct subtraction gives

$$j_E p_E = I - h\Lambda, \quad j_B p_B = I - \Lambda h. \quad (\text{PC26})$$

The chain equality follows as in PC18, since $\Lambda z = 0$. This proves the deformation retract with the newly retained Z in degree zero. It also proves the exact native-kernel sequence

$$0 \rightarrow Z \rightarrow K \xrightarrow{\pi_D} \ker \bar{\Lambda} \rightarrow 0. \quad (\text{PC27})$$

For surjectivity, choose x over c in $\ker \bar{\Lambda}$. Then Λx belongs to W , so choose v in V with $\Lambda v = \Lambda x$; $x - v$ belongs to K and projects to c . The kernel is exactly $K \cap V$.

The metric on PC25's source is the orthogonal direct sum of the inherited metric on Z and

$$Q_\pi + (h\Lambda L_\pi)^* G (h\Lambda L_\pi). \quad (\text{PC28})$$

Indeed, Z is orthogonal to L_π and to V_0 , hence to j_0 . The same determinant bound PC21 holds with $\min(r, q-d)$, and $\|h\| = 1/\sigma_r(T)$ when $r > 0$. There is no finite bound uniform through a changing positive rank unless this last positive singular value is bounded away from zero. The exact formulas still retain every dimension and metric on each rank stratum.

6. Two explicit observation behaviours at the same arithmetic collision

Take the auxiliary finite algebra $Q = y^2 + 1$, $D = y - z$ and coefficient metric $G=I$. Direct division gives

$$F_D = (z, 1)^\top, \quad B_D = -(1 + z^2), \quad \pi_D(a, b) = a - zb. \quad (\text{PC29})$$

For $\Lambda(a, b) = b$, $Q_B=1$, the observed fraction is $(1 + |z|^2)^{-1}$ and $h(t) = (zt, t)$. It remains defined at $z=i$, where F_D is a nonzero i -eigenvector of M . The rational vector has a pole but both the continued sector and its observed fraction remain finite and nonzero.

For the different observation $\Lambda(a, b) = a - ib$, $Q_B=1/2$ and

$$\frac{J}{H} = \frac{|z - i|^2}{2(1 + |z|^2)}, \quad h(t) = \frac{(z, 1)^\top t}{z - i} \quad (z \neq i). \quad (\text{PC30})$$

At $z=i$, $Z=V=K$ has dimension one, $W=0$, and $h=0$. Since $\pi_D=\Lambda$ there, $\bar{\Lambda}$ is the identity on the one-dimensional quotient. PC25 is exactly $[Z \oplus \mathbb{C} \rightarrow \mathbb{C}]$ with differential $(z, c) \mapsto c$, retaining the original one-dimensional kernel. These examples illustrate the proved maps; neither is represented as a numerical arithmetic-zero packet.

7. Receiving the actual proper-source comparison

For an independent proper-source pair retain its original matrices Q_N^p and $H_N^p = A_N - B_N^* C_N^{-1} B_N$, with the full target minimum in C_N . On explicitly fixed coefficient spaces of the same rank, set $\delta_{D,N} = \log \det H_{D,N}^E - \log \det H_{D,N}^B$. Direct cancellation, without identifying either proper metric with the reference, gives

$$\mathcal{R} \log \frac{\det H_N^p}{\det Q_N^p} = \mathcal{R} \left[\log \frac{\det H_N^p}{\det H_{D,N}^B} - \log \frac{\det Q_N^p}{\det H_{D,N}^E} \right] - \mathcal{R} \delta_{D,N}. \quad (\text{PC31})$$

Here $\mathcal{R}$ has the original cutoffs $q-1, q, 2q-1, 2q$ and signs $+, +, -, -$. The moving-family norm estimate controls the last term with the original angle and all cross terms. PC14 proves cancellation of its common resultant off the boundary; PC23-28 specifies the changed kernel when observation rank drops. PC31 leaves the two native comparisons as their actual unevaluated determinants. No arithmetic sign or initial kernel coefficient is inferred from this identity.

8. The exact kernel determinant on every observation-rank stratum

The boundary also has a determinant receiver, after retaining the common kernel Z explicitly. Give K/Z its full attained metric, and let $\tilde{\pi} : K/Z \rightarrow \ker \bar{\Lambda}$ be the isomorphism proved in PC27. The target inherits its metric from the entire quotient E/V . If $r = \text{rank } A$, then

$$\det(\tilde{\pi}^\dagger \tilde{\pi}) = \det^+(H^{-1/2} J H^{-1/2}). \quad (\text{PC32})$$

Here det-positive is the product of exactly the r positive eigenvalues, with the empty product equal to one. Its zero eigenvalues have not been discarded from the complex: their Z appears in PC25 and PC27.

To prove PC32, work in the Hilbert space $Z^\perp$, using the isometric minimum sections of the quotient by Z . The two subspaces are $K_0 = K \cap Z^\perp$ and $V_0 = V \cap Z^\perp$, with zero intersection. Choose orthonormal frames X, Y for them and put $C = X^\dagger Y$, using the Hilbert adjoint of these isometries. Orthogonal projection onto $V^\perp$ restricts to K_0 with Gram $I - CC^\dagger$. Projection onto $K^\perp$ restricts to V_0 with Gram $I - C^\dagger C$. The nonzero eigenvalues of $CC^\dagger$ and $C^\dagger C$ agree, as follows by the maps C and $C^\dagger$ on each positive eigenspace. Therefore their two identity-minus determinants agree. The latter determinant is exactly the product of the positive eigenvalues in PC32, since on $V = Z \oplus V_0$ the observation is zero on Z and has this Gram on V_0 . This proves the equation with both full minima retained.

In particular, if $\sigma_r(T)^2 \geq \ell > 0$, its logarithmic loss obeys

$$0 \leq -\log \det(\tilde{\pi}^\dagger \tilde{\pi}) \leq \min(\dim K - \dim Z, r) \log(1/\ell). \quad (\text{PC33})$$

The rank is bounded by the two dimensions of C . For $r=0$ the exact loss is zero. When $Z=0$ this is the earlier RF66 angle identity. At rank loss it is its exact continuation on K/Z , together with the retained inherited metric on Z . This gives the strongest finite determinant transfer at the boundary without an unproved positive-angle margin.

Finally the corrected section j_0 on $\ker \bar{\Lambda}$ maps isomorphically into K_0 . Indeed $\Lambda j_0 = 0$ by PC18 and j_0 is perpendicular to Z by PC28. It is the inverse minimum representative of $\tilde{\pi}$. Consequently

$$\log \det((j_0|_{\ker \bar{\Lambda}})^\dagger (j_0|_{\ker \bar{\Lambda}})) = -\log \det^+(H^{-1/2} J H^{-1/2}). \quad (\text{PC34})$$

Domain and target metrics are respectively the inherited metric from E/V and the inherited metric on K_0 . Thus, in the full-rank moving family, the chain-section correction on the actual kernel has error at most $2 \min(\dim K, d) E_{k,d,N}$, including growing $d = \lfloor q/\log^2(q+2) \rfloor$. Its four-cutoff return is $o(kq)$ by the complete activated bound. This larger-degree assertion follows from the restricted determinant identity, rather than applying the global source bound PC22 beyond its stated $d=O(k)$ range.

Human sources used in this continuation

Geneviève Dusson, Israel Michael Sigal and Benjamin Stamm, *The Feshbach-Schur map and perturbation theory* (2021), arXiv:2105.02058v1. The original-author TeX was read at lines 398–507, specifically `thm:isospF`, `QP` and `prop:U-prop-SA`. KG17–38 uses the exact elimination and reconstruction framework; the activated estimates are proved in this reader.

T. H. Koornwinder, R. Wong, R. Koekoek and R. F. Swarttouw, with W. P. Reinhardt, DLMF Chapter 18: 18.23.7 (Meixner-Pollaczek generating function) and 18.22.8 (recurrence). Their original TeX equations were reread for MF20, MF32 and MF42, through the full RF source bounds.

R. A. Askey and R. Roy, DLMF Chapter 5, 5.8.3 (Gamma product). Its retained original TeX supplies the source-envelope identity in MF5 and MF42 through IVO and RF. Complete versions, hashes, reading coverage and exact programme proof links are recorded in `SOURCE_LEDGER.json` and the accompanying rational-source ledger. A source reference records the stated use, not an exhaustive reading claim or a new proof of the earlier high-endpoint asymptotic.